June 2022 Paper A Recalls

1. Pregabalin MOA. Options – Calcium, NMDA, AMPA
    - potent ligand for the alpha-2-delta subunit of voltage-gated calcium channels
    - Pregabalin does not bind to plasma proteins and is excreted virtually unchanged by the kidneys
    - It is not subject to hepatic metabolism and does not induce or inhibit liver enzymes such as the cytochrome P450 system
2. A man unable to follow order - Ideational Apraxia

    an inability to follow a sequence of actions in the correct order e.g. Take a match out a box and strike it with your left hand

Apraxia

Apraxia is the inability to carry out learned voluntary movements (assuming everything works and they are not paralyzed).

The main types of apraxia are as follows:-

- **Limb kinetic apraxia** - an inability to make fine/ delicate movements
- **Ideomotor apraxia** - an inability to follow out learned tasks when given the necessary objects e.g. if given a hairbrush they try to write with it
- **Constructional apraxia** - an inability to copy a picture or combine parts of something to form a whole
- **Ideational apraxia** - an inability to follow a sequence of actions in the correct order e.g. Take a match out a box and strike it with your left hand
- **Oculomotor** - an inability to control eye movements

3. A man unable to use hammer, uses it later - Ideomotor Apraxia

    ↑ Above Q

4. Which neuronal cell that releases glutamate

    Astrocyte

5. Man had accident 5 years previously, now has bilateral coarse tremors and eye not focusing (reduced gaze) , sleeping tablets made his tremors better; which diagnosis -?PTSD, ?Adjustment disorder, ?Post concussional syndrome

    ? 5 years so No concussion PTSD less likely

6. Where can you find the Dentate gyrus? -
   Hippocampus > Episodic Memory

7. Receptor matching:
   Memantine Non-competitive NMDA antagonist (ketamine, phencyclidine also)
   Buprobion Norepinephrine-dopamine reuptake inhibitor (NDRI), and nicotinic acetylcholine receptor antagonist
   Naltrexone Reversible competitive antagonist at μ and κ receptors (δ receptor antagonist to a lesser extent)

8. Imprinting
   phase-sensitive critical attachment
   Lorenz
   Imprinting is rapid
   Unreinforced
   occurs during a clearly defined period (known as the critical period)
   Irreversible.

9. Neuroimaging findings in PTSD, , Neuronal axon damage, Antipsychotic effect

   PTSD Increased amygdala activation and smaller amygdala volumes are most consistently reported.

   Smaller hippocampal volume commonly appear repeatedly

   CJD

   vCJD Pulvinar Sign, Bilateral anterior basal ganglia high signal and thalamus

   Antipsychotic effect
   increased ventricular volume and reduced frontal and medial temporal lobe grey matter over time

10. Tetrabenazine receptor of highest affinity - ?specific
    Serotonin transporter (used in tardive dyskinesia, hyperkinetic disorders like tourette)
    VMAT-2* inhibitor, Reversible
11. Ketamine
    NMDA Antagonist

12. Form of therapy where ppl admit feeling helpless/powerless

Learned Helplessness , CBT

13. **EMI:** EEG Waves – Polyspikes,3-6 hertz, 2-4 hertz

3-6 hertz, Myoclonic

2-4 hertz Atypical Absence

14. Exposure of a patient to alcohol and drug misuse without actually taking the substances

Behavior therapy

15. Duloxetine use Liceneced for GAD, Pain disorders

16. A man whose friend died in a motor cycle accident an now has a paralyzed arm despite doctors not finding any neuro issues with the arm - ? Conversion disorder

17. Child waking up agitated

Sleep terrors are characterized by a sudden arousal from slow wave sleep with a piercing scream or cry, accompanied by autonomic and behavioural manifestations of intense fear.

Sleep terrors manifest as a severe autonomic discharge, with tachycardia, tachypnea, flushing of the skin, diaphoresis, mydriasis, decreased skin resistance, and increased muscle tone. The patient usually sits up in bed, is unresponsive to external stimuli, and, if awakened, is confused and disoriented. Amnesia for the episode occurs, although sometimes there are reports of fragments or very brief vivid dream images or hallucinations. The episode may be accompanied by incoherent vocalizations or micturition

18. The woman with multiple gynaecological, gastric pain…for 10 years , all tests normal, convinced drs missed something-? Delusional disorder, ? Somatisation

19. Area of brain affected in auditory hallucinations-

Language (Wernicke's area), and sensation of sound. Reduced gray matter volumes in the superior temporal gyrus (STG) have been reported in patients with schizophrenia

20. Area of brain involved in Default mode network

anatomically the **medial frontoparietal network** (**M-FPN**),

medial prefrontal cortex,
posterior cingulate cortex/precuneus
angular gyrus.

Active when a person is not focused on the outside world and the brain is at wakeful rest, such as during daydreaming and mind-wandering

Also Active thinking about themselves, remembering the past, and planning for the future

21. Example of activity that involves the use of default mode network

Above

22. Piaget, Kohlberg, Eisenberg, Erikson, Bowlby theories

Piaget

Piaget developed four stages of development that describe how children think and acquire knowledge.

Stage	age	Features
Sensorimotor	birth to 18-24 months	• Inborn motor and sensory reflexes • Primary Circular Reactions • Secondary Circular Reactions • Use of familiar means to obtain ends • Tertiary Circular Reactions and discovery through active experimentation • Insight and object permanence
Preoperational	2-7 years	• Deferred imitation • Symbolic play • Graphic imagery (drawing) • Mental imagery • Semiotic function (ability to use symbols) • Language
Concrete operational	7-11 years	• Egocentric thought replaced by operational thought (child now deals with information outside them) • Syllogistic reasoning (logical conclusions formed from two premises). • Conservation (Watanabe, 2017) of number (age 6-7), length (age 6-7), mass (age 7-8), area (age 8-9), weight (age 9-10), volume (age 11-12)

Koherlberg

Level	Age	Stage	Characteristics
Preconventional (level 1)	0-9 years	Obedience and punishment orientation	Focus is on direct consequence of action and unquestioning deference to power (not in terms of respect for an underlying moral order supported by punishment and authority)
		Self-interest orientation (aka Instrumental-relativist orientation)	Right behaviour defined purely by what is in the individual's own interest (you scratch my back and I'll scratch yours, and what is important is what I want)
Conventional (level 2)	9-20 years	Interpersonal accord and conformity (good boy - nice girl orientation)	Focus is on how the individual will appear to others Behaviour should accord with a consensus view on what is good.
		Authority and social order obedience driven (the law and order orientation)	What is lawful is judged to be morally right. Right behaviour is dictated by societal rules. There is a greater respect for social order and the need for laws.
Postconventional (level 3)	20 and onwards	Social contract (aka legalistic) orientation	Individual rights determine behaviour. The individual views laws and rules as flexible tools for improving human purposes. That is, given the right situation, there are exceptions to rules. When laws are not consistent with individual rights and the interests of the majority, they do not bring about good for people and alternatives should be considered.
		Universal ethical principles orientation	The right action is the one that is consistent with abstract reasoning using universal ethical principles

23. Locus coeruleus and the neurochemical pathway

Noradrenergic projection originates,at the locus coeruleus (pons) and ascends to most of the cortex via medial forebrain bundle.
noradrenergic projections also descend to the Spinal cord

24. Causes of Postural hypotension as a side effect and which receptor

Alfa one antagonist

25. Alpha 2 blockade

Agitation

Impotence

26. 5HT2a

Receptor	Action
$5HT_{1A}$	Antidepressant (agonist), anxiolytics (partial agonist)
$5HT_{1B}$	Aggression
$5HT_{1D}$	Antimigraine (antagonist)
$5HT_{2A}$	Antipsychotic (antagonist); hallucinogens (agonist / partial agonist); implicated in working memory; also seen in platelets and smooth muscles
$5HT_{2B}$	Stimulation may produce cardiac valvular fibrosis (dexfenfluramine)
$5HT_{2C}$	Anxiogenic and anorexic effect (agonists)
$5HT_{3}$	Antiemetic (antagonist)
$5HT_{6}$	Possible antipsychotic/antidepressant action (antagonism)
$5HT_{7}$	Regulation of circadian rhythm

27. Antidepressant with discontinuation syndrome risk – Venlafaxine?, Fluoxetine?
28. Agomelatine mode of action/receptor
    Agonist at melatonin MT1 and MT2 receptors and as an antagonist at 5HT2C receptors.
29. Which is extensively metabolized by CYP40
    CYP2D6

30. Hormone produced by the hypothalamus that regulates appetite – Neuropeptide Y,PPY,PPC,Cholecystokinin
31. Hormone produced by the adipose tissue that regulates appetite – Leptin, Neuropeptide Y

- **Neuropeptide Y** is produced by the hypothalamus and increases appetite
- **Leptin** is produced by adipose tissue and reduces appetite
- **Ghrelin** is produced mainly by the gut and increases appetite
- **Cholecystokinine** (CCK) is produced mainly by the gut and reduces appetite

32. Which pro-inflammatory interleukin is elevated in depression- IL 11?, IL 4?, **IL 6**

33. Altruism- ? innate, ?volunteering makes life better for old people

Vicarious but constructive and instinctually gratifying service to others.

It includes benign and constructive reaction formation, philanthropy, and well repaid service to others. Altruism differs from projection and acting out in that it provides real, not imaginary, benefit to others and from reaction formation that leaves the person using the defence at least partly gratified

34. MOA Cocaine, MDMA

Cocaine (and methylphenidate) directly blocks the dopamine transporter (DAT)

MDMA Blocks SERT Monoamine Transporter

35. Hospital management prioritizing early intervention unit for funding over the rehabilitation unit cos more pts use it. This is an example of which principle? – Autonomy?, Beneficence?, Non-maleficence?, Utilitarian

The greatest good for the greatest number.

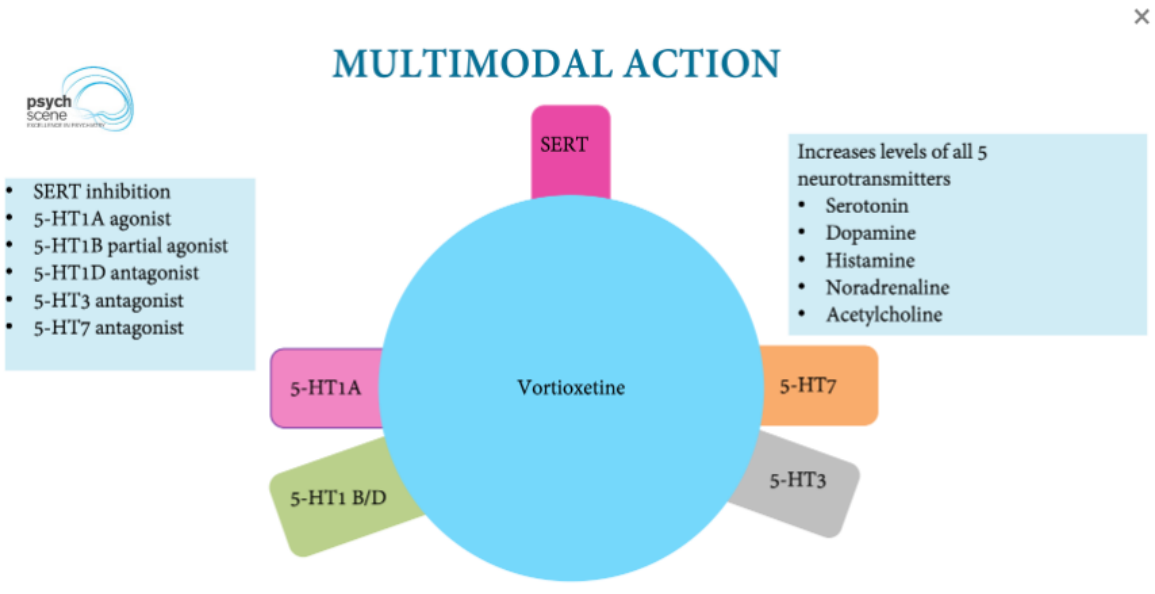
36. Vortioxetine receptors

5HT1A Full agonist

5HT1B partial agonist

Antagonist at

5HT3, 5HT7,5HT1D



- Sowa-Kućma M, Pańczyszyn-Trzewik P, Misztak P, et al. Vortioxetine: a review of the pharmacology and clinical profile of the novel antidepressant. Pharmacol Rep 2017; 69: 595–601
- Adamo D, Calabria E, Coppola N, Pecoraro G, Mignogna MD. Vortioxetine as a new frontier in the treatment of chronic neuropathic pain: a review and update. Ther Adv Psychopharmacol. 2021 Sep 3;11:20451253211034320.

37. Antidepressant for a patient with hyponatremia

Bupropion, Mirtazapine (SNRI, Mianserine as well)

38. Ion implicated in schizophrenia ?? (Calcium)

These alterations cause the most important inhibitory current in the brain, called $GABA_A$, to become excitatory. Inhibiting the transporter activity with bumetanide, an FDA-approved diuretic, rescued cognitive symptoms in mouse models of schizophrenia.
The polarity of the $GABA_A$-mediate current is determined by the balance of two chloride transporters: NKCC1 and KCC2

39. Brain organoid of pluripotent cell is at what age? what are the drawbacks?

6+ months ..

Drawback...lack of reproducibility

40. Area under the curve means? bioavailability

plasma concentration of a drug versus time

41. The graph on arousal and performance

The Yerkes-Dodson law is a model of the relationship between stress and task performance. It proposes that you reach your peak level of performance with an intermediate level of stress, or arousal.

42. Piagets assimilation, accommodation and equilibrium

Assimilation – The process of taking new information into an existing schema.

Accommodation – Altering a schema in view of additional information.

43. Differentiating vascular dementia from Alzheimer

Need Options

(step wise vs. progressive) vascular dementia pt is insightful.

44. Scale that has ‘no pain’ on one end and ‘excruciating pain ‘ on the other end - ? likert scale. ? Semantic ?Nominal ?Ordinal
Ordinal

45. Psilocybin receptor (magic mushroom)
5 HT2A

46. Purine properties (adenine, guanine)
Aromatic

47. Child language abilities at age 1,2,13 years

Stages of Language Development	age
1. Crying	Birth
2. Cooing	6 weeks
3. Babbling	6 months
4. Intonation patterns	8 months
5. One-word utterances	1 year
6. Two-words utterances	18 months
7. Inflectional words	2 years
8. Negative and Interrogative	2 ¼ years
9. Rare and complex constructions	5 years
10. Mature speech	10 years

Communication & Language Development Milestones 2-24 months

Milestone	Average Age of Attainment (months)	Developmental Implications
Smiles in response to face/voice	1.5	More active social participation
Babbles	6	Experimentation with sound
Inhibits to "no"	7	Response to tone
Follows one step command when said with gesture	7	Response to gesture
Follows one step verbal command when said without gesture	10	Verbal receptive language
Monosyllabic- mama/dada	10	Expressive language
Points to object	10	Interactive communication
Speaks first real word	12	Beginning of labeling
Speaks 4-6 words	15	Learns names of object and people
Speaks 10-15 words	18	Learns names of object and people
Makes 2 words sentences	19	Grammatization, 50+ words vocabulary

48. What not to take with fluoxetine - ? Aspirin, ?NSAIDs

NSAIDs: NICE guidelines advise 'do not normally offer SSRIs', but if given co-prescribe a proton pump inhibitor

Warfarin / heparin: NICE guidelines recommend avoiding SSRIs and considering mirtazapine
Aspirin: see above
Triptans: avoid SSRIs

49. Indicator for clozapine induced myocarditis (eosinophilia)

Myocarditis:

Myocarditis is rarely seen in clozapine use but carries with it a high mortality.

Studies have estimated the risk of clozapine-induced myocarditis to be 1 in 500 to 1 in 10,000 (Hass, 2007). The true risk is likely closer to the 1 in 500 rate, as often only a minority of adverse medication-related events are diagnosed and reported.

Most cases of myocarditis occur early in the course of clozapine treatment. The condition was detected within 16 days (median) of initiating clozapine in a study of 116 cases (Hass, 2007). Approximately 80 percent of cases of clozapine-induced myocarditis occur within four weeks of drug initiation, and 90 percent occur within eight weeks.

Postmortem examination has most commonly revealed damaged myocytes and eosinophilic infiltration suggesting a type I Ig E-mediated acute hypersensitivity reaction.

Some patients will have a mild and temporary increases in inflammatory markers and eosinophils at the beginning of treatment without progression to myocarditis.

Subsequent use of clozapine in cases with clear clozapine-induced myocarditis leads to recurrence of myocarditis in most cases when the drug is restarted.

Features include:-

Fever
Chest pain
Tachycardia
Dyspnoea
Flu-like symptoms
Elevated eosinophil count
Elevated cardiac enzyme levels

The gold standard for diagnosis of myocarditis is an endomyocardial biopsy, but this is not a practical initial approach.

Cardiac biomarkers, including creatine kinase and troponin, are routinely measured in patients with suspected myocarditis. Creatine kinase has been shown to be inferior to troponin for assessing myocardial injury, and is not useful for screening because of its low predictive value.

Outcomes of clozapine-induced myocarditis are poor, with mortality rates as high as 50%

50. Fetal malformation in Sodium Valproate (risk is 10%)

Neural tube defects (spina bifida 1%), facial dysmorphism, cleft lip and palate, craniostenosis (skull contractions), cardiac, renal and

urogenital defects, limb defects (including bilateral aplasia of the radius).

51. Which part of the frontal lobe is prominent in teenagers? Options

- Dorsomedial prefrontal
- Dorsolateral prefrontal
- Medial orbitofrontal
- Lateral orbitofrontal

NO ANS Found

52. EMI: When one's schema is corresponding with one's experiences. Options

- Accomodation
- Assimilation
- Equilibrium
- Organisation
- Centration

- Assimilation
  - Piaget defined assimilation as the cognitive process of fitting new information into existing cognitive schemas, perceptions, and understanding. Overall beliefs and understanding of the world do not change as a result of the new information.
  - Assimilation coccurs when the new experience is not very different form previous experiences of a particular object or situation we assimilate the new situation by adding information to a previous schema.
  - This means that when you are faced with new information, you make sense of this information by referring to information you already have (information processed and learned previously) and try to fit the new information into the information you already have.
  - For example, a 2-year-old child sees a man who is bald on top of his head and has long frizzy hair on the sides. To his father's horror, the toddler shouts "Clown, clown" (Siegler et al., 2003).
  - For example, a baby learns to pick up a rattle he or she will then use the same schema (grasping) to pick up other objects.

Accommodation

- Accommodation: when the new experience is very different from what we have encountered before we need to change our schemas in a very radical way or create a whole new schema.
- Psychologist Jean Piaget defined accommodation as the cognitive process of revising existing cognitive schemas, perceptions, and understanding so that new information can be incorporated. This happens when the existing schema (knowledge) does not work, and needs to be changed to deal with a new object or situation.
- In order to make sense of some new information, you actual adjust information you already have (schemas you already have, etc.) to make room for this new information.
- For example, a baby tries to use the same schema for grasping to pick up a very small object. It doesn't work. The baby then changes the schema by now using the forefinger and thumb to pick up the object.
- Also, a child may have a schema for birds (feathers, flying, etc.) and then they see a plane, which also flies, but would not fit into their bird schema.
- In the "clown" incident, the boy's father explained to his son that the man was not a clown and that even though his hair was like a clown's, he wasn't wearing a funny costume and wasn't doing silly things to make people laugh.
- With this new knowledge, the boy was able to change his schema of "clown" and make this idea fit better to a standard concept of "clown".

Equilibration

- Piaget believed that all human thought seeks order and is uncomfortable with contradictions and inconsistencies in knowledge structures. In other words, we seek 'equilibrium' in our cognitive structures.
- Equilibrium occurs when a child's schemas can deal with most new information through assimilation. However, an unpleasant state of disequilibrium occurs when new information cannot be fitted into existing schemas (assimilation).
- Piaget believed that cognitive development did not progress at a steady rate, but rather in leaps and bounds. Equilibration is the force which drives the learning process as we do not like to be frustrated and will seek to restore balance by mastering the new challenge (accommodation).
- Once the new information is acquired the process of assimilation with the new schema will continue until the next time we need to make an adjustment to it.
- **Equilibration** is a regulatory process that maintains a balance between assimilation and accommodation to facilitate cognitive growth. Think of it this way: We can't merely assimilate all the time; if we did, we would never learn any new concepts or principles.
- Everything new we encountered would just get put in the same few "slots" we already had. Neither can we accommodate all the time; if we did, everything

we encountered would seem new; there would be no recurring regularities in our world. We'd be exhausted by the mental effort!

53. Innate ability to gather information and integrate available info into a system of knowledge. Same options as Q52

As Above

54. Stimulated part of the brain in Visual hallucinations, Auditory hallucinations

Visual hallucinations were associated with reduced volume in bilateral occipital regions, right supramarginal gyrus and left fusiform gyrus, bilateral dorsolateral prefrontal cortex, frontal pole and the middle portion of the left cingulate gyrus.

Auditary Hallucination
left temporal lobe, in the peri-Sylvian region.

55. Progranulin gene- FTD

56. Confabulation . Options :

- Can only occur in Korsacoff syndrome
- May occur in clouded consciousness
- There is deliberate falsification of facts
- Autobiographical memory
- There is subjective awareness of memory impairment

Confabulation is a falsification of memory occurring in clear consciousness associated with organic states.
Suggestibility is a prominent feature of confabulation. It is often described in Korsakov syndrome. There
can either be confabulation of embarrassment or of fantastic nature

57. Tau positive and ubiquitin negative in FTD

Familial frontotemporal dementia with ubiquitin-positive, tau-negative inclusions

58. Which is not a risk for SSRI induced hyponatremia ? Options:

- Male sex
- Low body mass
- Low sodium level

Table 4: Risk factors for development of hyponatremia with psychotropics

Demographic variables: Increasing age, female gender
Physical characteristics: Low body weight
History: History of hyponatremia in the past
Baseline serum sodium levels: Low baseline levels
Environmental factors: Summer season
Nature of psychiatry disorder: Early-onset psychiatric illnesses, longer duration of psychiatric disorder, prolonged admission
Co-morbid medical conditions: Diabetes mellitus, hypertension, hypothyroidism, COPD, heart failure, head Injury, stroke, liver cirrhosis and malignancies
Concomitant medications: Diuretics, antihypertensives, antidepressants, cytochrome P450 inhibitors
Dose of psychotropic: Inconclusive (antidepressants and antipsychotics); higher dose in case of carbamazepine
Duration of treatment: during the initial phase of treatment

COPD: Chronic obstructive pulmonary disease

59. One of these is not a risk factor for delayed speech impairment. Options:
    - Immigration status
    - Maternal alcohol intake
    - Maternal smoking in pregnancy
    - Large families
    - Reduced cognitive stimulation

A number of risk factors are associated with delayed speech and language (Nelson 2006):-

- Family history
- Male gender
- Twins
- Lower maternal education
- Childhood illness
- Being born late in the family order
- Young mother at birth
- Low socioeconomic status

60. Which of these drugs do not cross the blood brain barrier? Options:
    - **Amisulpride**

- Olanzapine
- Quetaipine
- Clozapine
- Risperidone

61. Risk factors for Qtc prolongation. Options:
    - Male sex
    - Sinus tachycardia
    - **Hypokalemia**
62. Tardive dyskinesia- **D2 hypersensitivity ..** akathesia due to D2 occupancy.

63. Orthostatic hypotension- **Alpha 1 adrenergic blocker**
64. A cell was infected by toxins and inflammatory markers seen around the mileu. What process is this? Options:
    - Apoptosis
    - **Necrosis**
    - Entosis
    - Excitoxicity
65. Rate limiting step in alcohol metabolism. Options:
    - Acetaldehyde dehydrogenase
    - **Alcohol dehydrogenase**
    - Alcohol hydroxylase
66. How to reduce groupthink? - **open debate**
67. ADHD – **DRD4**
68. Ketamine MOA – NMDA **antagonist**
69. Cocaine receptor/MOA DAT

    The rewarding effect of cocaine is related to increase in dopamine concentration in limbic regions due to either dopamine transporter blockade or direct occupancy of D2 receptors. D1 is implicated in conditioned response to cocaine.
70. Southern blot- **specific DNA sequence**
71. Locum 1 hr extra **– fixed interval**
72. Gambling- **variable ratio**

71.Reversal of Caffeine inhibited by **Adenosine antagonist readily cross BB barrier , it is methyl xanthine derivative and completely reverses the effect of adenosine**

72.Thomas scale for infants. Options:

- Anger
- Impulsivity
- Control
- **Points from notes**
- **Activity**
- **rhythmicity**
- **approach/withdrawal**
- **adaptability**
- **intensity**
- **threshold**
- **mood**
- **distractibility**
- 9. attention span / persistence

73.Which is part of Maslow's original hierarchy of needs – Esteem?



Monotropy attachment to single figure mainly mother **According to Bowlby, the strong innate tendency to attach to one adult female is seen – this is monotropy**

74.Changes in pregnancy. Options:

- Increase in gastric PH
- **Increase in GFR**
- Increase in gastric motility

Pregnancy is associated with several **pharmacokinetic changes**:
✶ Delayed gastric emptying,
✶ Decreased GIT motility,
✶ Increased volume of distribution (5%),
✶ Decreased drug-binding capacity,
✶ Decreased albumin level
✶ Induced liver metabolic pathway,
✶ Increased GFR & renal clearance.

75. Monotropy attachment to single figure mainly mother **According to Bowlby, the strong innate tendency to attach to one adult female is seen – this is monotropy**

PAPER A RECALLS

DECEMBER 2021

1. Wave form in Delerium Teremens - **Hyperactive trace, fast**
2. Hormone secreted by hypothalamus that regulates hunger - **Neuropeptide Y**
3. Hormone secreted by adipocytes for regulation of food intake – **Leptin**
4. What reduces group think? **open debate**
5. A patient who thinks he has cancer despite negative clinical findings- **hypochondriasis**
6. Question on somatization disorder

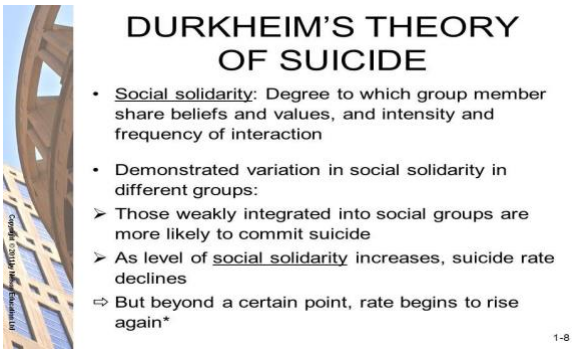
   Notes: excessive preoccupation and worry about somatic symptoms. It is more common in women and those with low education and limited incomes. The symptoms may (or may not) be associated with a medical condition.
   Note that DSM5 somatic symptom disorder evolved from somatization disorder and that in the former, the symptoms do not need to be medically unexplained to qualify for a diagnosis. The ICD-10 does require the symptoms to be medically unexplained.

7. Nucleotide most affected by methylation? **cytosine**
8. Nucleotide repeat in Huntington – **CAG**
9. Hypoplasia of part of brain in ASD – **cerebellar vermis**
10. What induces CYP1A2---**smoking**
11. What is the basis of modern day psychiatry diagnostic schedule---**operationalized**
12. Diagnostic criteria of delusional disorder....**no persistent hallucinations**

13. Persistence of sensation after cessation of stimuli ....**palinopsia (visual) palinaptia ?more correct**

    Palinaptia is the experience of tactile sensation outlasting the stimulus, so that an object held in the hand continues to be perceived well after it has been discarded.

14. Durkheim theory of suicide- Proposed theory of suicide, Until his work was published (le suicide) it was believed that suicide was an individual act only.
    "when social conditions fail to provide people with the necessary social goals and/or rules at the appropriate levels of intensity their socio-psychological health is impaired, and the most vulnerable among them commit suicide"



15. Buprenorphine - **Partial agonist at the mu-opioid receptor**

16. Vygotsky model of learning – **collaborative learning, scaffolding, zone of proximal development**

17. When does theory of the mind develop – **Autism, 4 years**

18. What causes distress in the child in strange situation experiment – separation anxiety (**Secure attachment and anxious resistant)**

19. Most common feature of Wernicke aphasia- **fluent/ receptive aphasia**, **impairment in auditory comprehension.**

20. Ideational apraxia- an **inability to follow a sequence of actions in the correct order e.g. Take a match out a box and strike it with your left hand**

21. Ideomotor apraxia- **an inability to follow out learned tasks when given the necessary objects e.g. if given a hairbrush they try to write with it**

22. 58-year old man who suffered a significant RTA, now has an inability to identify landmark near his home– **Visuospatial agnosia**

23. What is preserved in normal aging? **Procedural, episodic (remote), semantic**

24. Function of stroop test- **Response inhibition and set shifting**, **attention** bias

25. Alpha 2 autoreceptor,5ht2 and 5ht3 antagonist – **Mirtazapine?**

26. Antisocial personality Feature - **DLPFC dysfunction causing impulsivity**

27. Patient describes teaching hospital, as young doctors training place - **circumstantiality**

28. A disorder characterized by forgetting name, personal identity and episodic memory - **fugue**

29. D2 receptors seen in children before 5 years is greater than adult level and later regressed, this is called? - **Pruning**

30. Gaba B amino butyrate - **GHB – gamma hydroxybutyrate**, **dissociative anaesthetic**

31. DSM 5 has - **no subtype for schizophrenia**

32. Melanie Klein – **depressive position/Paranoid-schizoid position, SIPDOG**

33. Amygdala helps in **storing emotionally salient memories (SPMM – emotional consolidation of salient memories)**

34. What are the risk factors of prolonged QT – FEMALE GENDER, anorexia nervosa, hypokalaemia, magnesium and calcium, extreme exercise, extreme of age, use of antipsychotics, grapefruit juice, liver disease

35. Side effects of carbamazepine **Leucopenia, Thrombocytopaenia, Eosinophilia**

36. Most sensitive finding in clozapine induced myocarditis - **eosinophilia**

37. Postnatal depression
    EPDS – screening for postnatal depression
    HAMD/ MADRS – monitoring/ severity

38. Attitude Scales
    - Semantic differential scale = **Pairs of bipolar or opposite adjectives at either end of a series of scales eg weak/strong, ugly/beautiful. It's a 5 or 7-point scale.**
    - Likert scale (most popular) = **5 points scale with strongly agree to strongly disagree**

39. EMI
    Remembering what breakfast *is*? - **Semantic memory**
    Remembering what you had for breakfast – **Declarative memory**
    Remembering that after breakfast you took your bike out and cycled your usual route to work, and this felt tedious – **Episodic memory**
    Remembering how to eat breakfast – **Procedural memory**
    Remembering after you have eaten breakfast that you need to put up more cereal on your way home from work – **Prospective memory**

40. Ion most implicated in schizophrenia - ca?, k?, mg?...(**mg**). (**pt with schizo have lover erythrocyte magnesium levels than controls**) - Calcium activated **potassium channels, voltage gated sodium channel**

41. Anti-inflammatory cytokine involved in depression – il-4?, IL-6, Il-10
    Schizophrenia = IL-2

42. Component of ORIGINAL Maslow hierarchy of needs – **physiological needs, Safety needs, love & belonging, esteem needs, self-actualisation**





43. DSM-5 classification of eating where individuals don't take enough of what is required nutritionally, shows preference for enteral feeding and supplements - **avoidant/restrictive feeding disorder**

44. Not comfortable if not at the centre of attention - **histrionic personality**

45. Miserly with money and doesn't spend on self or others, saves for future purpose - **obsessive personality OCPD**

46. Inflated sense of self - **narcissistic**

47. ADHD mutated chromosome - **DRD4, dopamine receptor D4**

48. When does concrete operational begin – **7 years – 12 years**

49. Not a type of FTD – **Huntington**, Progressive non fluent aphasia, Pick's Disease, Semantic dementia, PSP.
(5 clinical subtypes of FTD: Behavioural variant FTD, primary progressive aphasia, progressive non fluent aphasia, semantic dementia, logopenic progressive aphasia)

50. Gambling - **variable ratio**

51. Locum hourly - **fixed interval**

52. Payment for consultation at a place where processing takes an unpredictable amount of time - **variable interval**

53. Identity moratorium question – **Has explored/is exploring but hasn't come to a decision**
Identity foreclosure question – **Has not explored but has come to a decision**

		Individual has committed to identity: Yes	Individual has committed to identity: No
Individual has explored identity options: Yes		Identity Achievement	Moratorium
Individual has explored identity options: No		Foreclosure	Identity Diffusion

54. Delay of gratification helps to measure - **Long term academic performance, social competence, good psychological health (Marshmallow experiment)**

55. Flumazenil helps to manage toxicity caused by – **benzodiazepine and zopiclone (Does not act barbiturates. Non- benzodiazepine agonist like zopiclone is also blocked by flumazenil).**

56. Neuroimaging used for measurement of neurotransmitters – **PET – measurement of regional brain metabolism by tagging organic compounds in metabolic pathway i.e. gives direct info about neuronal metabolism.**

MRS – dynamic measurement of resonance hydrogen and phosphate ions – odd number of protons
SPECT – measurement of receptor binding and distribution using radiolabelled receptor
fMRI – tasks-related changes in blood oxygenation level
Magnetoencephalography – measurement of static magnetic activity of brain

57. Images and symbols that represent human experience common to different cultures - **Archetype**

    A concept regarded as universally common to mankind; it contains various archetypes described by Jung - **Collective unconscious**
    The unconscious feminine aspect of male **– anima** (unconscious masculine aspect of female – animus)

58. Important kind of learning in the critical phase/period - **imprinting** (rapid, unreinforced, and irreversible) Lonrez

59. Drug with relative D2/D3 receptor activity (selective antagonist) – **Amisulpride** (Sulpiride – pure D2 antagonism)

60. Non-selective long-acting opioid antagonist – **Naltrexone** (short acting opioid antagonist)

61. What is likely observed by family members who show high expressed emotions to patient during interview - **critical comments** and hostility are the most predictive measures of relapse of schizophrenia.

62. What does Camberwell family interview (CFI) measures - **Expressed emotion in family.**

63. Question on family system theory viz -
    Enmeshment – no boundary in family
    Triangulation – in divorce, both parents want to be good to children, child sides with one patent creating enmeshment with one and withdrawal in the other
    Circular causality – bad habit affecting everybody in the family
    Detouring – like displacement, husband beating child because of conflict

64. Question on Erickson stages of psychosexual development - **Identity and role confusion can affect sexual orientation**

65. Which Hormone is expressed in the mother whose mother show good interaction with – **oxytocin**

66. To make a diagnosis of Bipolar 1 in DSM5 you need - **at least one episode of mania**

67. Neurochemical basis for tardive dyskinesia - **D2 supersensitivity**

68. Parkinsonism - high D2 occupancy in the striatum **(SPMM new questions: increased D2 sensitivity in the basal ganglia)**

69. Lyonization is associated with – **Klinefelter (XXY)**

70. Psychedelics MOA – **5HT2a partial agonism (and full)**

71. Drug absorption constant is independent of .food ,form of drug, extravascular route, elimination constant {From SPMM Notes – It is dependent on - Route of administration, chemical properties, form of the drug, rate of blood flow, solubility of drug (which depends on pH of drug size of particle in the formulation and pKa – ionization constant)}

72. Anti-hypertensive helpful for patient on lithium – **Atenolol; Furosemide** (Amiloride to reduce toxic effects; not hypertension)

73. Drug that follows nonlinear, kinetic at high dose – **fluoxetine, paroxetine,** and fluvoxamine

74. Patient at risk of hyponatraemia will benefit from - **mirtazapine**

75. Most important side effect of lithium - **worsening of psoriasis (nausea – most common; it also causes fine tremor, polyuria/polydipsia). Another question on spmm – most common = polyuria**

76. Topiramate - **weight loss; renal stones**

77. Why do we increase slowly when giving lamotrigine - **SJS**

78. Which Anti-epileptic is more implicated in aggression – **Levetiracetam** > topiramate; **vigabatrin**

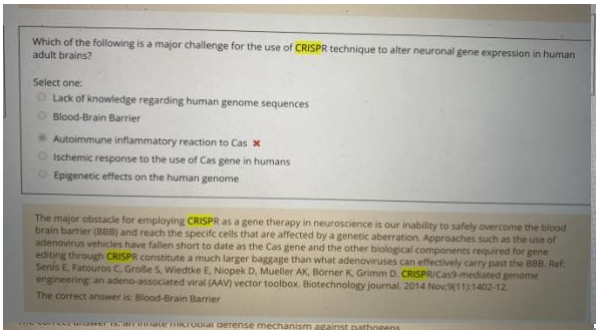
79. Empathy is principally mediated by...**DLPFC?** Hippocampus? Amygdala? (DLPFC and anterior insular cortex – google) **SPMM – supermarginal gyrus, especially on the right**

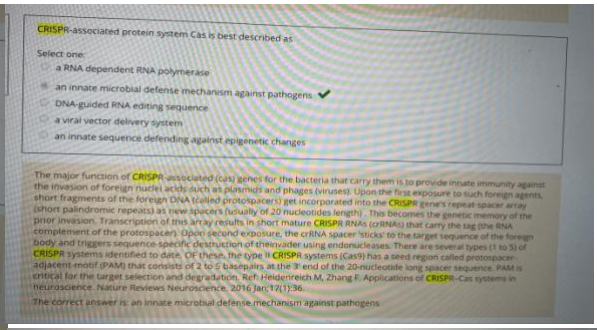
80. Question on association study – **what the exact disease-causing genes are.**
    Linkage analysis – location of culprit gene)

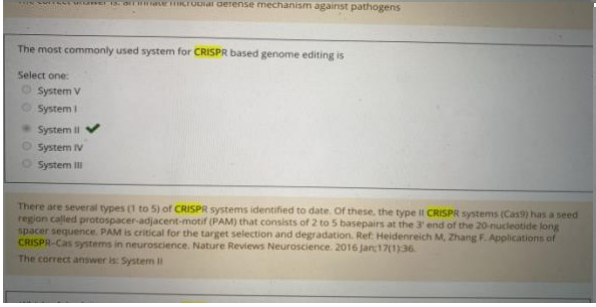
81. Area under curve in pharmacokinetics - **drug distribution or concentration in blood plasma as a function of time** (bioavailability in SPMM notes) – reflects the actual body exposure to drug after administration of a dose of the drug and is expressed in mg*h/L)

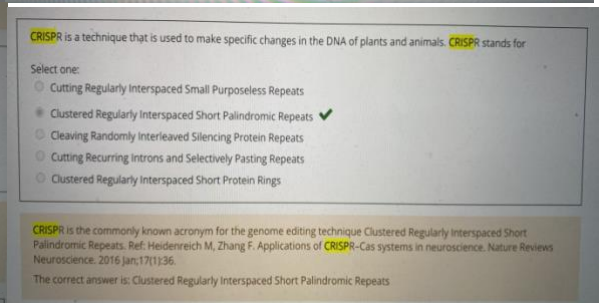
82. Question on first order kinetic...options were on clearance of the drug and their linearity – **Clearance is dose dependent; follows a linear curve.**

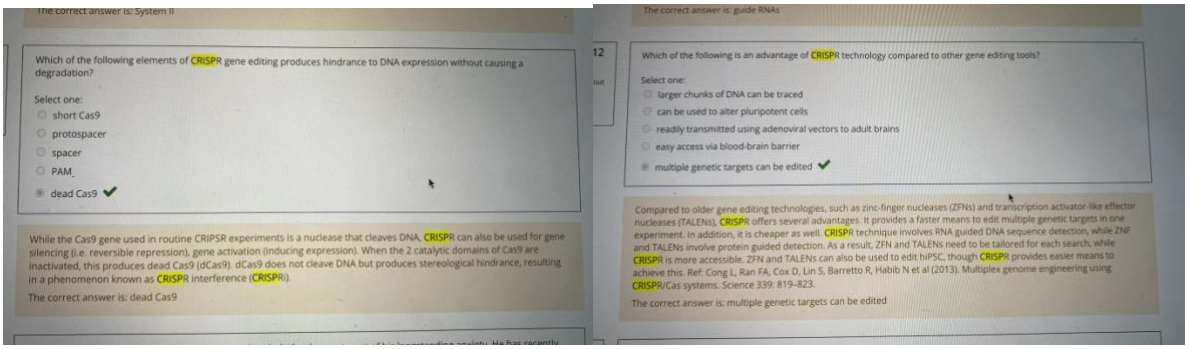
83. Question on CRISPR – Clustered Regularly Interspaced Small Palindromic Repeats.
    SPMM Questions on CRISPR include:
    - Major **challenge** for use to alter neuronal gene expression = **Blood Brain Barrier**
    - CRISPR-associated protein system Cas is best described as = **an innate microbial defence mechanism against pathogens**
    - The most commonly used system for CRISPR based genome editing is = **System II**
    - Elements of CRISPR gene editing that produce hindrance to DNA expression without causing a degradation = **dead Cas9**
    - Which of the following is an **advantage of CRISPR** technology compared to other gene editing tools = **multiple genetic targets can be edited.**
    - Nuclease based gene targeting techniques are = **CRISPR, ZFN, TALEN**











84. Question on pluripotent stem cell in the study of schizophrenia - **the draw back and what age group brain they resemble**

hiPSC – human induced pluripotent stem cells = have properties similar to embryonic stem cells ("brain grown in a dish")
Which mutations in Retts studied through hiPSC = **MECP2 Mutation**
Advantage of hiPSC over embryonic stem cells = **Ethically acceptable**
Homology directed repair = mechanism highly conserved across species and serves to repair damage to DNA.
gRNA refers to = guide RNAs
Endonucleases used for gene editing result in = double-strand break (DSB)
In scalp electromagnetic fields, what's most prominent source of recorded activity = post-synaptic electrical flow along the pyramidal dendrites
Technique requires a helium filled tank to enhance superconductivity = MEG (Magnetoencephalography)
True about MEG = signal is dominated by currents flowing tangential to the skull
Neuroscientific approach used to activate or inhibit specific neuronal projections in live animals using light exposure = Optogenetics
In Optogenetics, ChannelRhodopsin (pigment rhodopsin) = Blue light, Halorhodopsin and Archaerhodopsin = Yellow light.
Most common adverse effects of rTMS is = headaches (Scalp pain is commonest)
CLARITY is term used to describe = Hydrogel tissue clarification
DBS in resistant OCD targets = anterior limb of the internal capsule
DBS = pacemaker implanted under collarbone and clicked during visits.
In resting state fMRI, the frequency of BOLD signal captured = <0.1Hz
tDCS refers to = Transcranial direct current stimulation (modulate neural activity in brain in a non-invasive manner)
when applied to DLPFC, can reduce GABA to glutamate conc ratio = intermitten theta-burst stimulation.
Gyrification refers to development process of = **Cortical folding**
Completed by age 2 in normally developing children = **Cortical gyrification**
Functional connectivity = **refers to correlation in the time course of activation between 2 brain regions**
Frequency of transcranial magnetic stimulation effective and used in depression = **10 Hz (targeted at DLPFC as against Anterior cingulate cortex in DBS)**
Frequency of stimulation considered to have excitatory effect on neuronal cells = **5-20Hz**
Estimating subject specific motor threshold before prescribing tx dose required in **= TMS**
Clozapine-N-oxide bind to **= M3 Muscarinic**
Endogenous process used to regulate gene expression and to degrade foreign mRNA from phages **= RNA interference**

85. Diagram MOA
SERT and VMAT2 MOA
SERT – **Na/K**
ATPase
VMAT **– utilizes a protein gradient created by V-ATPases (**tetrabenazine and amphetamine – VMAT inhibitors)

86. How many turns do D2 receptors take in the cell – **7 transmembrane regions (G-protein coupled receptor)**

87. Neurasthenia
IC10 classification **- other neurotic disorders**

ICD 11 – **no more diagnosed, it has been replaced by bodily distress disorder**

88. Ganser syndrome ICD10 classification **- dissociative disorder**

89. Cue exposure - **conditioned**
    12steps – powerless, hopeless

90. Trisomy 18 - **Edward syndrome**

91. Rate limiting enzyme in alcohol metabolism **– alcohol dehydrogenase**

92. Precursor of serotonin - **5hydroxytryptophan**

93. What undergoes hydroxylation in the synthesis of dopamine **– L tyrosine**

94. Which receptor does ketamine act on **– Non-competitive NMDA antagonist**

95. Memantine moa **- NMA Antagonist**/ 5HT3 antagonist

96. Aura for frontal lobe epilepsy **- cephalic aura**

97. For TLE **- epigastric aura**

98. For parietal seizures – **somatosensory seizures**, **electrical sensation and tingling; numbness and tingling, heat, pressure or pain. (Jacksonian march)**

99. Drug used in ADHD that causes hypotension **– guanfacine/ clonidine** (atomoxetine – other nonstimulant ADHD drug)

100. Which area of the brain is stimulated during FMRI in
    - auditory hallucination – **primary auditory cortex (temporal lobe)**
    - visual hallucination – ventral occipital cortex

101. Test for executive function - **tower of London** (**problem solving and decision making) – deficit in planning**

102. Test for abstraction **– similarities**, **proverbs and cognitive estimates**

103. Question on parenting styles **- authoritarian, authoritative. Permissive**
    Authoritarian – military style
    Authoritative – firm, disciplined and loving
    Permissive
    Uninvolved/ Neglectful

104. Gestalt laws – **symmetry and order, proximity, continuity, closure, common fate, similarity**

105. Seeing a face in the sky – **pareidolia (illusion)**

106. Choosing group therapy over individual therapy because it will help a larger group - **Utilitarianism**

107. Withdrawal of a drug from the market because it causes cardiac event **- non maleficence**

108. Drug with the lowest tendency to cause hyperprolactinemia - **Aripripazole**

109. Basis for prions disease - **Protein misfolding**

110. Question on reciprocal inhibition

111. ECT neurochemical findings - reduces (ach?glutamate? Gaba?n acetylaspartate?) – **reduction in serotonin 5HT2 receptor density.** This has antidepressant action. ECT also enhance dopaminergic neurotransmission in the brain. Differential BDNF expression in different areas of the brain is believed to be responsible for antidepressant action. ECT also reduces cortisol levels to normal among patients with moderate depression in due course.
(Through bdnf - serotonin receptors, global decrease in presynaptic serotonin 5HT1A receptors)

112. Question on hemineglect - the definition
Unawareness and unresponsiveness to objects, people or things on one side of the body (non-dominant parietal lobe lesions –dressing apraxia, constructional apraxia, visuospatial agnosia, anosognosia)

113. Attention cause conversion from **sensory to short term memory (**Rehearsal – from STM to LTM)

114. Neuronal cell that aids transfer of glutamate **– astrocytes (can release glutamate, atp, nitric oxide and prostanglandins)**

115. Neuronal cell that aid transmission through GABAergic neurons in the caudate **- medium spiny neurons/ spiny projection neurons**

116. High performing student takes alcohol be able to do a presentation in public **- social phobia**

117. Adjustment disorder scenario

118. Question on Holme and Rahe life event scale **–** death of spouse (100), **divorce (73)**, marital separation (65), death of close family or jail time (63)

119. Man who witnessed a friend's demise 6weeks ago is unable to move limb despite absence of neurological deficit **- conversion**

120. Question which had this option...behaviourism, mentalization Empathy

121. Valproate risk of causing congenital anomalies**. - 8-12% (10%)**
Risk of developmental delay and other cognitive difficulties in infants = 31-40%

122. Losses own culture at the expense of imbibing another's - **assimilation**

123. Causes of qt prolongation diuretic use? **– hypokalaemia, hypomagnesemia, hypocalcaemia, extreme exertion, extremes of age, female gender, anorexia nervosa.**

JUNE 2021

1. Delayed gratification in a child is associated with?
A. **Higher academic achievement**, lower tendency towards frustration and aggression later in life

2. What reduces group think/polarization
A. **Open debate**

3. Duloxetine is used in?
A. Peripheral neuropathy (60mg – 120mg)

4. Which one is reward pathway?
A. **Mesolimbic**
B. Mesocortical (motivation and emotion)

5. EMI: reading others state of mind is?
A. Mentalisation – ability to understand mental state of oneself and others that underlies behaviour

6. Autism shows abnormality of which structure?
A. Hypoplasia of cerebral vermis (? **Reduction in size of cerebellum**)

SPMM: **Also Decrease in volume of corpus callosum**, Increase caudate/amygdala

7. Familial FTD – autosomal dominant
A. Tau –ve, Ubiquitine +ve

8. Which one is Autosomal Recessive?
A. Wilson's

9. Picture of pre-degree
A. Autosomal dominant?



Autosomal dominant / autosomal recessive / X linked recessive / X linked dominant

10. Wernicke's is what kind of Aphasia?
A. Fluent or Receptive Aphasia (Broadmann's 22)

11. A patient cannot perform a complex task like using hammer but can perform each individual step?
A. Ideational apraxia

- **Ideomotor apraxia** - an inability to follow out learned tasks when given the necessary objects e.g. if given a hairbrush they try to write with it
- **Ideational apraxia** - an inability to follow a sequence of actions in the correct order e.g. Take a match out a box and strike it with your left hand

12. Tyrosine hydroxylase converts
A. L-tyrosine to L-Dopa

13. Which gene is associated with ADHD
A. DRD4

14. Anti-social Personality is associated with?
A. highest heritability (38%)???
Note: low arousal theory – ADHD and Antisocial PD seek self-stimulation by excessive activity in order to transcend their state of abnormally low arousal.
SPMM Q: Which PD has the strongest support for a genetic condition?
Dependant, Schizoid, Anankastic, Antisocial, **Schizotypal**

15. Not an ICD-10 criteria for Persistent delusional disorder?
A. No auditory hallucinations
B. **Duration of symptoms of 2 weeks.**
C. Non -bizarre delusion
{Note: minimum of 3 months, non-bizarre delusions, relatively preserved functionality,

16. Multifactorial in origin?
A. Autism
B. Fragile X
C. Down
D. Turner's syndrome

17. Parental figure is important in shaping the attachment behaviour in children.
A. Bowlby Attachment theory

18. EMI: This theory involves the stages of development. A person moves to next stage after first is complete, (chronological phase-specific psychosocial crisis)
A. Erickson's theory (psychosocial)

19. EMI: main theme here is obedience, punishment, praise, and universal ethics
A. Kohlberg's theory (moral)
Preconventional 0-9 – obedience and punishment, self-interest orientation (instrumental-relativist)
Conventional 9-20 – good boy-nice girl, authority and social order obedience driven (law and order)
Postconventional 20above– social contract (legalistic), universal ethical principles

20. QTC prolongation occurs with?
A. Haloperidol
B. Quetiapine
C. Risperidone

21. Innate behavior of a species is called - **Imprinting**

22. EMI: 13-year-old child can speak- **20,000 words**
18 months old child speaks- **40 words**
24 months – **telegraphic, grammatically pairs words, vocab more than 240words**

23. EMI: 36 months child - **can understand grammatical syntax accurately**
4 years**- correct use of grammar**
5 years- **language akin to adult speech**

24. Trail making test is for –
A. **Motor perceptual speed,**
B. Selective Attention
C. Verbal memory
D. Vocabulary

25. Stoop test is used for - A. **Executive function (response inhibition and set shifting)/ selective attention**

26. During interview a doctor offered hand shake to a schizophrenic patient. Patient move forward and then withdraw hand for 20 times - **Ambitendency**

27. Locus coeruleus contains which neurotransmitter - **Noradrenaline**
Ach – Nucleus of Meynert
Serotonin – Medial and dorsal raphe nuclei
GABA – Nucleus accumbens
Dopamine – Substantia nigra

28. EMI: What to check before starting the Methylphenidate - **Blood pressure**

29. EMI: What to check before starting the Atomoxetine, agomelatine - **LFTs**

30. More than 50% of delta wave seen in which part of sleep - **NREM 4**

31. EMI: Which aura occurs in occipital seizures - **Visual illusion**

32. EMI: Which aura occurs in Parietal lobe seizures - **Somatosensory**

33. EMI: Taupathy? **A. FTD with Picks, Alzheimer's, Progressive supranuclear palsy, Corticobasal degeneration, Dementia Pugilistica**

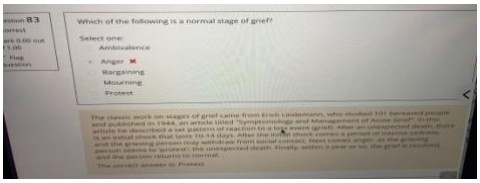
34. EMI: **Neuregulin NRG1 (on chromosome 8)** - **Schizophrenia**

35. Changes in pregnancy –
A. Increase GFR
B. decrease PH
(others: increased gastric pH – decrease in production of gastric acid, delayed gastric emptying, Decreased GI motility, Increased volume of distribution (5%), decreased drug binding capacity, decreased albumin levels, induced liver metabolic pathway, increased drug renal clearance, increased hepatic/renal blood flow)
(In Pregnancy, only **APE decrease** = Absorption, Protein binding, Emptying)
(In Old age - **FTPT increases** = Fat, Free drugs, Toxicity, pH, T-half life (others reduce, liver metabolism unaffected)

SPMM –
Decreased GFR
Increase in gastric pH
Increased gastric emptying
Increased protein binding of medication
Decreased volume of distribution

36. Which one is a screening tool?
A. EPDS
B. MARDS
C. Hamilton

37. Which one is normal stage of grief?
A. Protest
B. Mourning
C. Anger
D. Bargaining
E. Ambivalence



Lindemann stages of grief: Shock, sadness, protest and resolution.
Bolwby: Shock-numbness, Yearning-searching, Disorganization-Despair, Reorganization-Resolution/Repair
Kluber Ross: DABDA (Denial Anger Bargaining Despair Acceptance)

38. Where is Broca's area located?
A. Inferior frontal lobe
B. Superior frontal lobe
C. Temporal lobe
D. Parietal Lobe
E. Limbic area

39. Which kind of memory loss in schizophrenia?
A. Working memory
B. Procedural memory
C. Delayed memory

40. EMI: What are the MRI changes in patient with Alzheimer? **A. Medial temporal lobe atrophy** (hippocampus) Entorhinal cortex – first to atrophy

41. Aripiprazole mechanism of action - **A. Partial D2 agonist and 5 HT2A antagonist**

42. EMI: Vortioxetine mechanism of action?
A. 5HT1A agonism (also has 5HT3 Antagonism effect)

43. EMI: Half-life of Fluoxetine? **A. 4-5 days**

44. EMI: Half-life of escitalopram? **A. 30 hours**

45. What is true of fundamental attribution error? **A. More common in western world**

46. Which part of brain develop last? **A. Frontal lobe**

47. Changes that occurs during maturation of adolescent brain? **A. Increase in pruning and synapsis**
(decrease in grey matter and increase in white matter)

48. People use to organize current knowledge, images and provide a framework/match for future understanding?
A. Schema
B. Personal construct
C. recognition
D. Recall

49. Person is unable to recognize the place? landmark **A. Visuospatial agnosia**

50. EMI: Patient has visited 5 hospitals at different times and all test done are normal. He is not accepting that everything is normal and convinced that he has cancer? A. **Delusional disorder –** (*Hypochondriasis)*

51. EMI: Patient has multisystem complaints. All labs are normal. He is convinced that doctors have missed something-grievous disease. **A. hypochondriasis**

52. EMI: Hormone secreted by posterior pituitary? **A. Oxytocin** (and Vasopressin/ADH)
Anterior pituitary – GH, ACTH, TSH, Prolactin, LH, FSH

Hypothalamus – Somastotatin, PIF, Releasing factors

53. EMI: Hormone secreted by Liver? **A. Insulin growth factor**

54. EMI: Diazepam mechanism of action? A. **GABA A receptor agonist**

55. Propanalol mechanism of action?
C. B1+B2 adrenergic antagonist (non-selective, competitive)

56. ApoE4 heterozygous increases risk of Alzheimer's by?
A. 3 – 4 times
Homozygous – 10-30 times (10-15 times)

57. EMI: Drug withdrawal from market due to cardiac side effects? Beauchamp and Childress
A. Non-Maleficence (do no harm)
Beneficence (will act in a way that will be helpful to patient).

59. EMI: A Rogerian psychotherapy developed by incorporating the Beck's analytic therapy?
A. ***Cognitive therapy ???***
B. Behaviour therapy ???
C. Analytic therapy
D. DBT
E. Rational emotive therapy
F. Group therapy
G. Cue therapy
H. 12-step program
I. Alcohol Anonyms

60. EMI: During substance abuse therapy, Alcohol or substance signifier is used in controlled conditions?
A. Cognitive therapy
B. Behavior therapy
C. Analytic therapy
D. DBT
E. Rational emotive therapy
F. Group therapy
G. Cue therapy ???
H. 12-step program
I. Alcohol Anonyms

61. EMI: During substance abuse treatment, people believe that they are helpless, hopeless and powerless?
H. 12-step program

62. A child protest and cries if mother goes but becomes comfortable once mother returns?
A. Secure attachment
B. Insecure attachment
C. Disorganize attachment

63. Durkheim proposed that suicide is more common in?
A. Socially deprive people?
B. unmarried lonely couple
C. Living in urban area living for the long time

64. EMI: An adolescent whose is alcoholic drinker. He is hesitant to go out with parents. He fears that he may vomit at party.
A. Social phobia
B. Agoraphobia
C. Alcohol dependence

65. EMI: A Son lost his father in road traffic accident. Crying unconsolably, What is diagnosis? A**. Acute stress reaction**

66. EMI: side effect of topiramate **A. Weight loss**

67. EMI: clozapine side effect - **A. M4 *agonism*** - hypersalivation

68. EMI: Carbamazepine side effect? **Leucopenia**
A. Agranulocytosis *(with clozapine use)*

69. What is difference of parkinsonian tremor from cerebellar tremor?
A. 2-4 Hz **– *Parkinson (4-6Hz)***
B. They are intentional – *Cerebellum (2-3HZs)*

70. What is true regarding schizophrenia DSM 5 classification?
A. Subclasses retained
B. No categories

71. FMRI is used for?
A. Functional location of brain tissue
B. Bony lesion
C. Calcified lesion

72. EMI: Atonic seizure
A. ~~?No change~~ **(Generalised spike and wave is typical with atonia at the time of the slow wave)**

73. EMI: Atypical absence Seizers
A. **(slow <2.5Hz generalised spike and waves)**
Typical absence Seizures - 3Hz

74. Drug of choice in a patient who has Bipolar affective is taking lithium and has diagnosis of hypertension. Drug of choice would be.
A. Atenolol
B. Frusemide
C. Acetazolamide
D. Thiazide diuretics

75. EMI: A person is confused about his role and he is in ***mid-adulthood*** crises?
A. Stagnation vs generativity

76. EMI: Increase startle response occurs in?
A.
B. hyperekplexia
C. alyxthymia
D. Blunted

77. EMI: Cannabinoid receptor? **A. Anandamide receptor**

78. A rat is placed in a cage. He is given electric shocks at some area of cage whenever rat goes to that area. After sometimes, rat start avoiding that area.
A. Avoidance learning
B. Operant learning
C. Classical conditioning

79. Conventional stage of development-Q17 **(Kohberg)**
Piaget - Concrete operational

80. Theory of mind is test? – also **Sally Anne**
A. Reading minds in the eyes
B. PENN
C. Eysenck's

81. Which one is Gestalt principle? **A. Proximity**

82. EMI: During old age a person give up due to Socio- economic reasons
A. Cost benefit analysis??
B. Reengagement
C. Re arrangement

SPMM Note:
Old age – theories of role change:
1. Social disengagement theory: **Mutual withdrawal of society and the individual** occurs; increased individuality and shrinking life space are inevitable, moves towards death.
2. Social (non) reengagement theory (aka activity loss theory): Ageist society reduces the social interaction that older adults can have; **withdrawal is not mutual but forced.**
3. Social exchange theory: Age robs people of the ability to engage in reciprocal roles; retirement is a special social contract wherein **productivity is exchanged for increased leisure and decreased responsibilities.**
4. Socio-emotional selectivity theory: **Wise investment of social energy in old age is to limit social interaction to those who are most familiar.**

83. Which of following is not seen in DNA epigenetics?
A. Phosphorylation
B. DNA Acetylation
C. DNA methylation
D. Histone methylation
E. Histone Acetylation ???

(SPMM Q: Which of the following is not an epigenetic mechanism?
A. Glycation
B. DNA Methylation
C. Histone methylation
D. Histone phosphorylation
E. DNA Acetylation)

Epigenetic mechanism most associated with development of **schizophrenia = DNA Methylation (SPMM Q)**

84. Layonisation occurs in?
A. Klinefelter's syndrome
B. Down
C. Prader Willi
D. Fragile X

85. Which one is need?
A. self Esteem
B. Food
C. Love

86. Which one is Metabotropic Receptor?
A. CB1 CB2 Endocannabinoid
B. GABA-A
C. Glycine
D. AMPA

E. Kainate

87. Which drug we cannot give with fluoxetine A. Salbutamol ????

88. Neurasthenia is classified under which category in ICD-10 **A. other neurotic disorders**

89. What is the diagnostic criteria for bipolar 1 in DSM 5? **A. One manic episode**

90. What percentage get metabolic syndrome in CATIE trial? – **prevalence 40.9%, Men 36%, women 51.6%**
A. 40-45%
B. 30-35%
C. 50-55%
D. 20-25%
E. 60-65%

91. what is half life of GHB
A. 30-60 minutes
B. 12 Hours
C. 24 Hours

92. EMI: Lithium **A. Worsening of Psoriasis**

93. A female is afraid of heights is shown images of sky-high building and standing on top of building in imagination **A. Implosion**

94. A personal is afraid of spiders is taken to a room gradually containing spiders along with relaxation. A**. Systemic desensitization**

95. Evaluation of pleasure valence occurs in which part of brain?
A. Amygdala
B. Cingulate cortex
C. Hippocampus
D. DLPC
E. Ventral pallidum

96. EMI: This enzyme diffuse the substrate at rate of its availability.
A. Acetylcholine acetrate
B. Adenylate cyclase
C. Aldehyde dehydroginase
D. Beta Hydroxylase
E. COMT
F. GABA
G. MOA
H. Alcohol dehydrogenase

97. EMI: This enzyme is not present at birth.
A. Acetylcholine acetrate
B. Adenylate cyclase
C. Aldehyde dehydroginase
D. Beta Hydroxylase
E. COMT
F. GABA
G. MOA
H. Alcohol dehydroginase

98. Rogerian therapy develop on the basis of Beck's cognitive distortions is? Person-centered therapy

A. Analytics B. Behavioural C. Cognitive therapy D. DBT E. Cue base F. Alcohol anonymous G. Rational emotive H. 12 steps

During Alcohol and substance abuse treatment, alcohol or substance abuse signifier is used under control condition?

A. Analytics B. Behavioural C. Cognitive therapy D. DBT E. Cue base F. Alcohol anonymous G. Rational emotive H. 12 steps

In this therapy group work together in sense that they are helpless and powerless?

A. Analytics B. Behavioural C. Cognitive therapy D. DBT E. Cue base F. Alcohol anonymous G. Rational emotive H. 12 steps

98. A chronic alcoholic is telling stories with no link. What one occurs in confabulation?
A. Clouding of consciousness
B. Autobiographical memory impairment

SPMM: Kosarkoff hallmark = anterograde amnesia (dorsomedial thalamus)

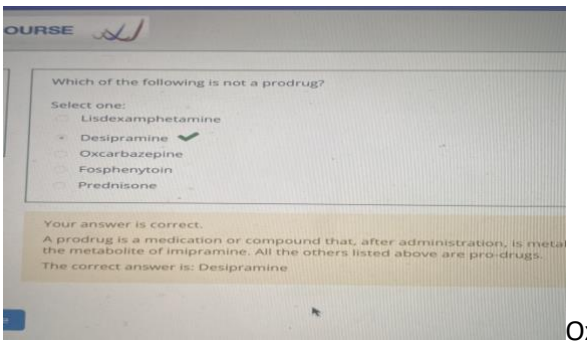
99. What about question on optogenetics.
A. Single is introduce with single beam of light.
B. Single gene introduce with multiple light beams.
C. Multiple genes introduce with single light beam.
D. Multiple genes introduce with multiple light beams + Something missing in all option-parallel or vertical.
E. One option had variant copy

100. Philosophy of modern classification system is based on??
A. Operational system

101. One question was social theory on cost benefit analysis.
A. Socio-economic
B. social exchange theory
C. Social deprivation theory
D. Social re engagement

101. Attitude, cooperation, and approach during the interview for the MSE assessment is called? **Rapport**

102. Which one is not a prodrug.
A. Desipramine (Active metabolite of Imipramine)
B. Lisdexamfetamine (A prodrug = metabolite is Dexamfetamine)
C. Oxcarbazepine
D. Fosphenytoin
E. Prednisolone



Oxazepam (metabolite)

103. This process is in form of imaginative mental activity that lets us perceive and interpret human behaviour in terms of intentional mental states.
A. Mentalization
B. Personal construct
C. Mindfulness

104. Reciprocal inhibition in phobias happens in?
A. Operent condition
B. Classical conditioning
C. Systematic desensitisation

105. EMI: This helps for release of serotonin neurotransmitters from vesicles. (Calcium channel)
A. 7 membrane, G protein
B. Adenylate cyclase
C. ATP base ion channel
D. Ligand ion channel
E. Ligand ion sodium channel
F. Ligand ion potassium channel
G. Ligand ion chloride channel
H. Ionic sodium and calcium channel ???
I. Inositol channel

106. EMI: Cannabinoid receptor
A. 7 membrane, G protein

107. EMI: This is for mechanism of lithium.
I. Inositol phosphate

108. EMI: What receptor directly reverses effect of caffeine?
(Caffeine is Adenosine antagonist)
A. 7 membrane, G protein
A. Adenosine.....
B. Adenylate cyclase
C. ATP base ion channel
D. Ligand ion channel
E. Ligand ion sodium channel
F. Ligand ion potassium channel
G. Ligand ion chloride channel
H. Ionic sodium and calcium channel
I. Inositol channel
J. GABA A facilitator
K. GABA B facilitator
L. NAMDA
M. AMPA

107. excessive startled response occurs?
A. Hyperepklepsia
B. Paramimia
C. parathymia.
E. anastrophy,
F. paramimia,
G. parathymia
Anastrophy is the 2nd stage of delusion formation (after trema)
H. apophenia,

108. EMI: Inability to express emotions **A. Alexthymia**

109. EMI: reduce and straight effect. Limited emotions **A. Blunted effect**

110. What is percentage of birth defect if a patient continue Valproic during pregnancy? **8-10%**
A. 3-4%
B. 30-40%

111. Stimulus sensation remains long after removal of contact?
Palinaptia
(visual persistence – palinopsia)

112. Situation and psychological interpretation
A. Anantrophe B. trema

113. serotonin precursor
A. L-tryptophan

114. The function of MRS –MRS measures
A. metabolites
B. proton flow ??

115. Young Girl asks mom if she will a baby brother?
A. Formal operational

116. The big 5 factor theory suggest that personality is? (Note: NEO decrease with age)
A. Can change over time
B. independent of self

117. Which drug has a non-linear pharmacokinetic?
A. fluoxetine

118. Hypomanic patient's diagnostic feature ICD or DSM?
A. functionally remains intact
B. precursor to recovery from mania ?

119. Ganser syndrome is?
A. Dissociative state

120. The part of brain for empathy
A. DLPFC ???? Supramarginal gyrus
B. Amygdala
C. Insula

121. The most common site of methylation? – **Cytosine carbon 5???**
A. Adenine???
B. uracil
C. thymine

122. Heritability is defined as?
A. Genetic contribution to phenotypic variation in population
B. Genetic contribution of disease in an individual
C. Degree of penetrance of a disease
Psychmentor: **Heritability is the proportion of phenotypic variance attributable to genetic variance.**

A. Phenotype in Population

123. Which of the following is precipitating factor for mental health illness
A. bereavement

124. Patient with normal tone, left with his arms folded
A. waxy flexibility
Posturing - maintaining the same posture for long periods. A classic example is the 'crucifix'. An extreme version of posturing is catalepsy

Waxy flexibility (cerea flexibilitas) - patient can be positioned in uncomfortable postures, which are maintained for a considerable period of time.

125. Which of them degrade dopamine in presynaptic
A. MAO (MAO B more?)
B. Acetylcholine esterase

126. Which of the following is an amino acid
A. Glutamate
B. enkephlin
C. substance P

127. EMI: The fibres that carry efferent from the cortex
A. pyramidal cells ---- corticospinal tract

SPMM: Cell types using glutamate as a NT = **Granular and Pyramidal cells**
Lines cerebral ventricles = **Ependymal cells**

128. EMI: Which cells for myelination in the CNS
A. oligodendrocytes
Myelination to PNS = **Schwann cells**
Cells that transport glutamate to promote clearance from synapse = **Astrocytes**
Cells which line the ventricles of the brain and the central canal of spinal cord = **Ependymal cells**

129. Man who immigrated and lost his culture
A. assimilation
B. integration

130. Southern blot is used for – (SNoW-DRoP)
A. sq of DNA

131. EMI: In which is there a direct current applied using saline soaked pads?
A. ECT
B. rTMS
C. DBS
D. Transcranial direct current stimulation

132. EMI: In which of them are the electrodes implanted in the brain?
A. Deep Brain stimulation
B. ECT
C. rTMS
D. Vagus nerve stimulation
E. Transcranial direct current stimulation

133. Which describes loss of identity formation
A. Identity diffusion????
B. foreclosure
C. moratorium

		Individual has committed to identity: Yes	Individual has committed to identity: No
Individual has explored identity options	Yes	Identity Achievement	Moratorium
	No	Foreclosure	Identity Diffusion

134. Which drug is safe for pt with hyponatremia?

A. Amitryptaline
B. Mirtazapine .. moclobemide .. bupropion
C. venlafaxine
D. Citalopram
E. Fluoxetine

135. EMI: Pro-opiomelanocortin (POMC) is precursor of which hormone in the brain?
A. ACTH
B. enkephlin
C. substance
D. Enkapheline
ACTH is a 39 amino acid peptide derived from a larger precursor protein, proopiomelanocortin (POMC). This precursor molecule is synthesized principally in the corticotropic cells of the anterior pituitary and in the arcuate nucleus of the hypothalamus

Some interesting new terms:
Parathymia an abnormal, contrary, or inappropriate mood, as seen in schizophrenia
Alloesthesia is a condition in which a sensory stimulus, given on one side of the body, is perceived to be at the corresponding area on the opposite side.
Incongruence between action and gesture – paramimia? – a misuse of gestures in expressing thoughts or words that produce an appearance of inappropriateness of affect, Parathymia? – incongruency btw affectivity and context paramimia (incongruent affect thing), flattened affect (according to sims)

Hypomania – mental efficiency

Mechanism of action of vmat 2 – **integral membrane ATP dependent**, serotonin transporter – **passive sodium or sodium dependent**

Carbamazepine toxicity – ?blood dyscrasia ?hair loss
Confusion, drowsiness, ataxia, dysmetria, nystagmus, myoclonus, seizure, coma = neurologic
Cardiovascular = Tachycardia, Hypotension, Wide QRS

Caution with fluoxetine – naproxen
MAOIs, Triptans, TCAs, fentanyl, lithium, tramadol, buspirone, amphetamine, St johns wort

Caffeine – adenosine antagonist

Melanin klein paranoid schizoid – splitting, depressive - ambivalence

Propranolol – non selective b1 b2 b3

Diazepam – GABA A agonist/facilitator
Empathy - ?insula ?amygdala (spmm new questions: **supramarginal gyrus** – inferior parietal lobe - particularly on the right appears the most implicated in Empathy)

Reward pathway - **mesolimbic**

Ability to perceive communication
No eeg change – auto dom frontal lobe epilepsy (interictal eeg often normal, ictal eeg unhelpful)

Not in grief theory – agitation

Language production in the brain – **Broca's in inferior frontal**

Wernickes = Superior temporal?

Pleasure valence – ?**orbitofrontal cortex**

Multiple symptoms – somatization

Metabolic – PET (not MRS)

Activated in excess diffusion of substance? **Acetylcholinesterase**

Social exchange theory – cost effective

Not affected by ageing – VIPO

Durkheim – **social effect on individual for suicide**

Dysbindin – **Schizophrenia**

- DTNBP1 (dysbindin) - chromosome 6 (6p 22.3)
- COMT - chromosome 22
- NRG1(neuregulin 1) - chromosome 8 (8p 21-22)

Genetic question about cross section of cohort study - I think the question was around population studies at one specific point in time

Last to mature – anterior part of frontal lobe

Cell which makes efferent connections in brain

Oligodendrocytes – myelin sheats in CNS

GABA effect? Intracellular Ligand gated permeable to chloride channel

Heritability – genetic attribution on oneself

Reciprocal communication - ?rapport

Improving adherence in EMI – cognitive based intervention, behaviour modification, motivation and compliance

Not a prodrug – desipramine

Carbamazepine most common adverse effect

Aripiprazole – partial d2 agonist

SSRI to be taken with caution – Naproxen

DSM V classification of schizo

Awkward position for long – waxy flexibility ?

Drawing Rogerian indirect counselling with Beck's CBT

I think motivational interview

Confabulation – autobio memory? deliberately filling gaps - falsify with insight about memory deficit?

SPMM: Confabulation is a falsification of memory occurring in clear consciousness associated with organic states. Suggestibility is a prominent feature of confabulation. It is often described in Korsakov syndrome. There can either be confabulation of embarrassment or of fantastic nature. “honest lying”
In pseudologia fantastica, there is fluent plausible lying (falsification of memory), with the statements made extreme and of grandiose nature. Is usually associated with dissocial or histrionic personality disorders.

Preserved in old age – **A. Procedural memory** B. working memory C. Spatial awareness D. Short term memory E. Retrograde memory

HiY Clinical Examination 088
If you perform a routine cognitive examination on a 90-year-old woman with no history suggestive of dementia, which of the following is least likely to be seen?

Select one:
- Reduced new learning efficiency
- Reduced working memory
- Reduced visuospatial ability
- Reduced motor speed
- Reduced attentional span

Reduced short-term, working memory is a feature of dementia. It is not expected as a part of healthy aging. But reduced motor speed, visuospatial skills, attentional span, and decreased new learning ability are seen in healthy aging process.

The correct answer is: Reduced working memory

Saline electrodes with electrical therapy - TDCS

Cells that make efferent connections in the brain – pyramidal cells

Ganser syndrome – dissociative disorder

Question on good mood – euphoria

There was a question on klaus conrad's theory - ?apopheny, trema, anastrophe = What was the question?

Stage	Term
I	Trema (derived from Greek, colloquial for stage fright) Meaning Delusional mood (or atmosphere) Characteristics Undefinable, but increasingly upsetting quality spreads from salient aspects to entire perceptual field. Patient feels anticipatory excitement, suspiciousness, alienation, fear, guilt, depression, or combination of these. Patient may perform abrupt, seemingly meaningless actions
II	Apophany (Greek apo [away from] + phaenein to show → revelation) Meaning Delusion as revelation (Aha-Erlebnis) Characteristics Perceptual Gestalt experienced incompletely in terms of its expressive rather than its objective material holistic qualities. Inability to transcend current perspective or to shift frame of references. Abnormal connectedness between seemingly unrelated meanings. Delusional perception, misidentification. Relentless ("monotonous, repetitive") spreading of the delusion as both "elastic" and fixed to new gestalts.[7,18,26] Progression of delusions from external to inner "space", delusional body sensations. Patient uncritically receptive and unable to detach, as if trapped between sleeping and waking. Thought insertion, thought broadcasting, hallucinations
III	Anastrophe (Greek, ana- (back) + strephein (to turn) → turning back) Meaning Patient feels self to be passive middle point (subject-directed complement to world-directed apophany) Characteristics Delusions of reference. Events and perceptions are related to self

Note: Conrad's later stages apocalyptic-catatonic, consolidation (or partial remission), and residual defect state are not presented here.

MOA of Duloxetine – SNRI not in option

Highest QTC prolongation... Haloperidol

camberwell Assessment of Need (CANE) = for elderly... coming to think of it, Critical comments was another question

Which factor is most influential according to Eysenck theory **– biological basis** to personality

Big five one was again which among these are correct - I chose traits change over time or something

Emotional valence= Amygdala or Cingulate? – **Research – anterior cingulate cortex (also spmm – implicated in depression emotion modulation)**

Precipitating factor for depression – bereavement

Also, can you recall if there was a clinical question stem mentioning an individual with head injury, now non- fluent speech...lesion is in?: Answer was inferior frontal lobe(since-Broca's)....

There was another one. Self centred, needs,needs approval – Erikson theory (Eisenberg moral theory)

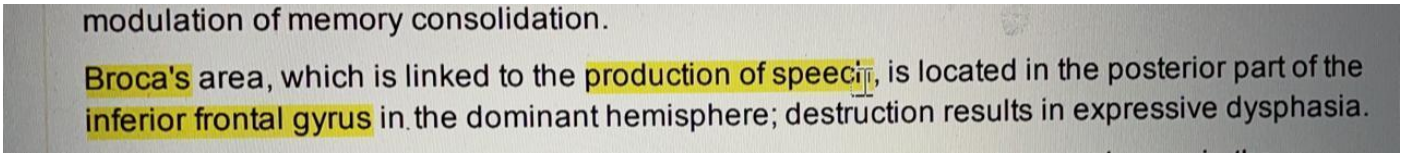
There was another one. Regarding. Basic Instinct needs that has to be fullfilled before moving to next.was is erikson too? No Freud for sure, related to Freud phallic stage or some other (Freud???)

Another one was which of the following true about melanie klein theory - Answer was paranoid schizoid position happens in 1st yr (?3months) ?denial

What age concrete stage begins was a question too – 7-11yrs (Piaget)

Oxyhaemoglobin was Fmri

Language production or Language processing



What can be used in hyponatremia? Mirtazipine

Drug whose metabolite is an active metabolite.... sertraline

5 year olds start understanding passive speech.

And 13 year olds have 20k vocab

Bowlby attachment theory – innate tendency to form attachments, bonding concerns mothers feelings for infant and differs from attachment.

Stage	Period	Description
Preattachment	Birth to 6 weeks	Baby orientates towards mother, follows them with their eyes, and turns to mother's voice. Complete attachment not yet complete and baby is still comfortable with unfamiliar people
Attachment in the making	6 weeks to 6-8 months	Infant becomes more attached to one or more persons in the environment. Begins to show different reactions to familiar people versus strangers. Separation anxiety not yet present
Clear cut attachment	6-8 months to 18-24 months	Infant cries and shows distress when separated from the caregiver (separation anxiety). This phase can occur as early as 3 months. On being returned to the caregiver the crying stops. Child shows preference for selective caregivers and will seek to be near them
Formation of reciprocal attachment	18 months to 2 years and on	Child begins to understand that their caregiver may leave and return so separation anxiety lessens. Child may be less physically clingy but may begin to find more intelligent ways to influence their caregiver to get their needs met (bribing and sulking for instance)

Freud - **Structural theory**

In his structural model, Freud proposed the existence of three separate areas:-

- Id
- Ego
- Super ego

The Id is the area that contains the instinctive drives. It operates under 'primary process thinking', acts according to the 'pleasure principle', and is without a sense of time.

The ego attempts to modify the drives from the Id with external reality. It operates on the 'reality principle'. It has aspects that are conscious, preconscious, and unconscious. It is home to the defences mechanisms.

The super ego constantly observes a person and acts as critical agency. Freud claimed it developed from internalised values of a child's main carers. The 'ego ideal' is part of the super ego and represents ideal attitudes and behaviour. It is useful to think of the super ego as the conscience.

Freud introduced his **topographical model** of the mind in his book *the interpretation of dreams*. In this model he divided the mind into three regions:-

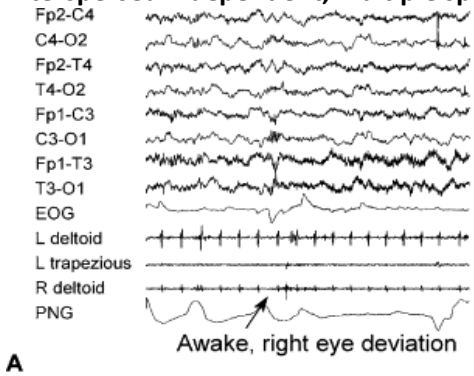
- The conscious system
- The preconscious system
- The unconscious system

Xanomeline underwent phase 3 trials as a treatment option for schizophrenia. It acts **via M1/M4 agonism** . Xanomeline showed a trend toward improving cognitive function in Alzheimer's, and produced robust and dose-dependent reductions in psychotic symptoms in AD. In schizophrenia, early trials showed efficacy in both the positive and the negative symptoms along with improvements in verbal learning and short-term memory but, a high degree of **gastrointestinal side effects were observed.**

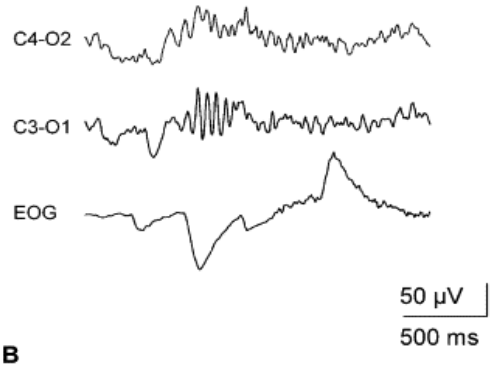
Dec 2020

1. **Hypsarythmia**

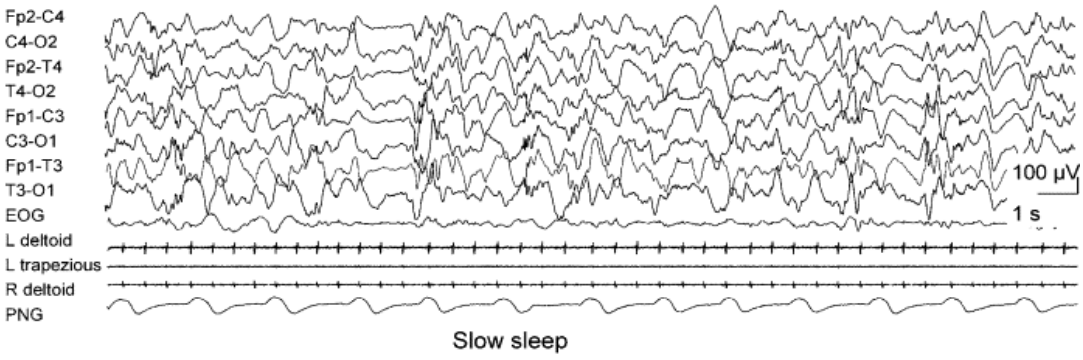
-Hypsarrhythmia is defined as **continuous, high-amplitude, arrhythmic, asynchronous delta activities with interspersed independent, multiple spikes**. From: EEG Pearls, 2006.



A



B



C

-Hypsarrhythmia is the most common interictal EEG pattern associated with **infantile spasms**. The most common clinical description is a sudden, symmetric, tonic muscle contraction producing flexion/extension of the trunk and extremities, although a variety of movement patterns have been described.

-**West syndrome** is a constellation of symptoms characterized by epileptic/infantile spasms, abnormal brain wave patterns called hypsarrhythmia and intellectual disability.

2. Function of MRS

-magnetic resonance spectroscopy (MRS) is typically described as the only noninvasive technique that can reliably quantify in vivo concentration levels of key metabolites, including glutamate and γ-aminobutyric acid (GABA)

3. Wilson chromosome

- The gene for Wilson's disease **(ATP7B) was mapped to chromosome 13**.
- Wilson's disease is inherited as **an autosomal recessive**
- Wilson disease is a rare genetic disorder characterized by excess copper stored in various body tissues, particularly the liver, brain, and corneas of the eyes. The disease is progressive and, if left untreated, it may cause liver (hepatic) disease, central nervous system dysfunction, and death.

4. Azheimer – tau

5. Ftd

6. Lewy body dementia -

7. 3 ques on identity theory etc

Erikson's observations about identity were extended by Marcia, who described four identity statuses:

		Individual has committed to identity: Yes	Individual has committed to identity: No
Individual has explored identity options	Yes	Identity Achievement	Moratorium
	No	Foreclosure	Identity Diffusion

8. gender schema theory
- Gender schema theory states that children actively construct mental representations about that which defines males and females by observing individuals in the culture in which they live.

10. Brain default network.
-**The default mode network -** state of **wakeful rest (MAPP)** = **Medial prefrontal cortex, Angular gyrus, Precuneus, Posterior cingulate cortex PCC = Midline cortical structures**

Dorsal attention network or Central Executive Network or frontoparietal (FLID) = Frontal gyrus(middle), Lateral superior parietal lobe, Intraparietal sulcus, Dorsolateral prefrontal cortex

Salient Network (FADA) – Frontal operculum, Anterior insula, Dorsal anterior cingulate cortex, Anterior prefrontal cortex, with some subcortical structures like thalamus and putamen.

11. Gait for
Huntington – chaotic
upper motor and – scissoring
lower motor – foot drop, high stepping
cerebellar disease – chaotic, broad based
parkison – festinate, shuffling
normal pressure hydrocephalus – magnetic
pregnancy -

12. Psilobin (magic mushroom) with MOA – 5HT2A agonist
- Cross-tolerance between the different hallucinogens has been demonstrated, and research shows a common mechanism of action through serotonergic (5-HT) pathways. Psilocybin is a strong agonist at 5-HT2A as well as a moderate agonist at 5-HT1A and 5-HT2C

13. Contagion theory;
-Contagion theory is a theory of collective behavior which explains that the crowd can cause a hypnotic impact on individuals. ... These entire theories share a basic idea of how individual can act briefly unreasonable or inconsistent within a group and become normal when they are not in it.
If a people interested in country music goes to a rock concert along with friends and the friends are enjoying very much, the person will start to enjoy too due to group feeling and mass contagion. The person forgets his/her own likes and dislikes to follow the group behavior.

CROWD behavior: **(Contagion – contagious, convergent – industrialization, emergent – new trend)**
three different theories to explain crowd behavior: contagion theory, convergent theory, and emergent norm theory.
-The convergence theory is the one which postulates that all the societies as they move from the early industrial development to complete industrialization tend to move towards a condition of similarity in terms of the general societal and technological norms.
Some examples of convergence theory include Russia and Vietnam, formerly purely communist countries that have eased away from strict communist doctrines as the economies in other countries, such as the United States, have burgeoned.
- Emergent norm theory hypothesizes that nontraditional behavior (such as that associated with collective action) develops in crowds as a result of the emergence of new behavioral norms in response to a precipitating crisis.
The ALS ice bucket challenge is an example of collective behavior that raised money towards medical research

14. **Pontine lesion** – locked in lesion, quadriplegic, eyelid blinking…….
(Left Pontine – left face (ipsi), right hemiparesis (contra), Left Midbrain – right face and hemiparesis the same, but left CN 3 palsy - ipsi)

15. **Deindividuation** - Deindividuation is a concept in social psychology that is generally thought of as the loss of self-awareness in groups. Theories of deindividuation propose that it is a psychological state of decreased self-evaluation

and a decreased evaluation of apprehension causing abnormal collective behaviour, such as violent crowds and lynch mobs. Conversely it also explains peoples tendency to donate more readily to charity when in groups.

16. **John bowlby stage of grief** except bargaining
Normal Grief: 4 phases
Shock-numbness, Yearning-searching, Disorganization-Despair, Reorganization-Resolution/Repair
Kluber Ross: DABDA
Erich Lindemaan: Shock, Sadness, Anger/Protest, Resolution

17. selective d2 antagonist - **Sulpride**

18. **Sally anne test** - TOM
- The Sally–Anne test is a psychological test, used in developmental psychology to measure a person's social cognitive ability to attribute false beliefs to others.
- The child passes the test if he or she knows that Sally will look for the item in the basket and not the box. Typical children struggle with this test as 3-year-olds, but most pass it by 5 years of age.

Reading of the mind in the eye test – Baron Cohen 1997

19. EEG seen as normal, what disease? Beginning of AD? Delerium tremens?in between seizures – inter-ictal

20. EEG centrotemporal spikes – Benign Rolandi Epilepsy
- EEG (electroencephalogram): Children with CECTs have spikes on their EEG in the centrotemporal regions of the brain. These findings help confirm the diagnosis. MRI (magnetic resonance imaging): This test is usually normal in children with CECTS. It is often not needed if the history and EEG are very typical.(**Childhood epilepsy with centrotemporal spikes -CECTS)**

21. over the counter medication cannot take with MAOI - decongestant

22. ADHD medication cause hypotension – **Clonidine**, Guanfracin, Atomoxetine (non-stimulant)
Methylphenidate – Hypertension (other stimulants are Dexamfetamine, Lisdexamfetamine)

24. emi receptor :
Zopiclone – Gaba A,
Corticosteroid -

25. emi development how many word:
1 year old = 2-3 words
13 years old = 20,000 words

26. anorexia purging type finding: **metabolic acidosis** (vomiting – metabolic alkalosis)

27. Wernicke aphasia : **fluent**?

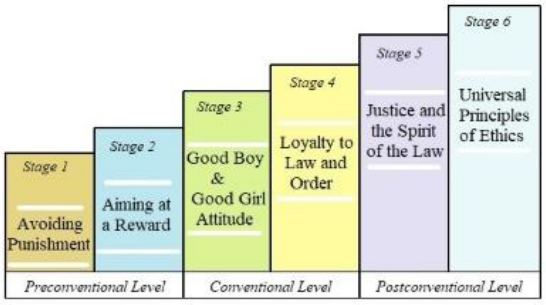
28. Human research: Helsinki – identifiable human data (ethics – Nuremberg, Declaration of Malta – hunger strike, Declaration of Geneva = revision of the Hippocratic oath, Tokyo – not to participate in torture of prisoners or detainees, Lisbon – international statement of rights of pts, Ottawa – principles for optimal child health)

29. See cloud as face: **pareidolia**

30. Fundamental attribution error: **focus on other people's behavior.**
Self-serving bias – is a type of attribution error

31 attachments not distress and ignore – anxious avoidant

32. kohlberg moral



33. computational psychiatry: reward prediction error is difference between received reward and? **What is predicted**

34. Neurotransmitters with fast excitatory: **glutamate**

35. Antipsychotic increase absorption with food? **Lurasidone**
Benzo – without food (food reduces their absorption)
Asenapine – buccal

36. Apoe4 heterozygotes, how many fold? **3-4times**
Homozygous – 10-15 times

37. Pregnancy physiological changes - **increase GFR, increase renal elimination** (Only **APE decrease** = absorption, protein binding, emptying)
Old age - **FTFT increases** = fat, toxicity, free drugs, T-half life (others reduce, liver metabolism unaffected)

38. at training hospital, ask patient where are you? Answered “at young doctor training place”
Neologism? Tangential? Verbigeration? Verbreiden? **Circumstantiality**

39.landmark agnosia?
Visuospatial agnosia

December 2019

SSRI questions = post MI, switch to mania = buproprion, fluovoxamine, sertraline,
Half life citalopram 30hrs,

group psychology =

EMI
LBD = alpha synuclein, ubiquitin
FTD = tau protein

Extrovert **decreased RAS**

Huntington's Gene location – **Chromosome 4, 4p16.3**

Delayed gratification is the ability to postpone an immediate gain in favor of greater and later reward

Group polarization refers to the tendency for a group to make decisions that are more extreme than the initial inclination of its members.

Hypertension **side effect of Venlafaxine.**

Duloxetine is Serotonin Noradrenaline Reuptake inhibitor-**SNRI**

Both olanzapine and clozapine, drugs with a high risk of weight gain, bind strongly to the **histamine H1 and serotonin 5-HT2C receptors.**

Reward pathway - **Mesolimbic**

Lurasidone should be **taken with food** (at least 350 calories). Administration with food substantially increases the absorption of Lurasidone

Dorso lateral Pre frontal cortex used in - ? Mentalization, execution, working memory

Introversion with **high Reticular activating system**

Autism shows **atrophy of cerebellar vermis** on scan
SPMM: Also **Decrease in volume of corpus callosum**, Increase caudate/amygdala

Which is not a type of Familial Frontotemporal Dementia? ?psp or ?huntington

Highest risk of seizures - **Clozapine**

LBD - DAT Scan (differentiate

Antipsychotic with the lowest risk of hypotension - **Aripiprazole**

Cause of weight gain with antipsychotics - **5HT2c antagonism**

Equivalence of **10mg diazepam - 1mg of lorazepam, 25mg chlordiazepoxide, 10mg Nitrazepam, Cloba Ox T (COT 20mg)**

How many times does **dopamine receptor cross the cell membrane**? 7 – seven membrane

Delay of gratification as measured in childhood is associated with?
(**Academic achievement**)

Hypnotic suggestibility is associated with - ?personality traits

Glutamate decarboxylase is involved in synthesis of? - **GABA**

Disorders of consciousness: Global impairment of consciousness, disturbed sleep-wake cycle (my answer: delirium)

Constant, dream-like perceptions - oneroid state, catatonic schizo, twilight

ADHD associated with which gene - DRD4

Personality disorders: "Consistent irresponsibility, as indicated by repeated failure to sustain consistent work behavior or honor financial obligations." (my answer: antisocial personality disorder)

"Appears indifferent to the praise or criticism of others." (my answer: schizoid personality disorder)

"Transient, stress-related paranoid ideation or severe dissociative symptoms." (my answer: Borderline personality disorder)

There was one which I think was narcissistic personality disorder.

Which of the following is most common in human brain (among D1, D2, D3, D4, D5) – Commonest dopamine receptor in the brain - D1

There were questions on "which of the following is a metabotropic receptor?" - CB1

Differences in mentalisation between adolescents and adults?

Which of these is (? underactive) in autism spectrum disorder – amgydala, insula, orbitoparietal or parietotemporal SPMM: **Also Decrease in volume of corpus callosum**, cingulate gyrus. Increase caudate/amygdala

What type of receptor do corticosteroids act on in the brain? ??Nuclear receptors

What type of receptor does (?zopiclone/?zolpidem) act on? Omega, GABA - A

Questions on attachment theory

Questions on language developmental milestones =
Vocalizing alone - cooing 2 months, babbling 6months (?6 to 9 months)
Using two-word sentences (?20 – 24 months)

Questions on symptoms such as feeling of limbs twisting = ?paraschemazia (body image distortion)

When doctors offers a handshake, patient offers a handshake, then withdraws hand, then offers hand, then withdraws it, repeatedly, over 20 times = ambitendency

Which neurotransmitter is contained in the raphe nuclei = serotonin

Patient on depot antipsychotic (as well as an antidepressant) complains of back stiffness, takes 10 minutes to get out of bed in the morning = parkinsonism, axial rigidity

Question on which antipsychotic is least likely to cause postural hypotension = aripiprazole

Which antipsychotic is least metabolised in the liver? Sulpride

Antipsychotic most likely to precipitate epileptic seizures (well, something along those lines) Clozapine

What tests to do before starting methylphenidate = BP

What to check before starting atomoxetine, agomelatine= LFTs

Which of these is not a variant of frontotemporal dementia? ?progressive supranuclear palsy

Genes linked with FTD
- The majority of the heritability of FTD is accounted for by autosomal dominant mutations in three genes: progranulin (GRN), microtubule-associated protein tau (MAPT- The MAPT gene is found on chromosome 17) and chromosome 9 open reading frame 72 (C9orf72)

According to ÆSOP study, what accounts for the nine-fold increased risk of schizophrenia in patients of African-Caribbean descent? = immigration

Also questions on EEG findings.

Questions on pharmacology - changes in distribution of drugs in older adults, changes in pregnancy, etc.

What's screening scale for depression?
EPDS, Hamilton Depression scale, MRDS

What's Not grief stages?
Protest, Shock and Numbness , Yearning and Searching , Disorganization and Despair, Reorganization and Recovery

EEG findings of Delreium Tremens
increase in beta fast activity

Q sort is used for = personality traits = research methodology

Raphi Nuclei (Serotonin)

Auditory hallucination – temporal lobe

depression in elderly - deep white matter hyperintensities, (lofepramine and nortriptyline preferred for elderly)

pulvinar nucleus -???pulvina sign

impairment working memory = schizophrenia, dlpfc

least to be affected in old age – cranial nerve nucleic, inferior nucleus to be chosen first

person throws stones - deindividuziation,

alzeihmer - medial temporal lobe atrophy on MRI (entorhinal first to happen)

lewy bodies dementia eeg option - REM behavior sleep disorder

least hepatotoxic = asenapine
- Asenapine, sold under the brand name Saphris among others, is an atypical antipsychotic medication used to treat schizophrenia and acute mania associated with bipolar disorder. It was chemically derived via altering the chemical structure of the tetracyclic (atypical) antidepressant, mianserin.

Least hypotension - aripiprazole

seizure risk increased - clozapine,

more than 50 hrs - ?flupenthixol

wife hitting husband on bed - akathisia, restless

girl with clonus or hyperreflexia-serotonin syndrome,

success of person internal attribute,

temperament poorly stable over developmental years,

weschlers intellectual ability,

gender schema, = gender identity, gender constancy

psychoanalytic =

cognitive development theory = Piaget = sensorimotor, preoperational,

social preferencing,

daily activities – MOHOST, ?**vineland social maturity scale**,

person performs well in presence of other = hawthorn

duration criteria for somatization = 2 yrs ICD10, 2 months DSM

panic disorder = pregabalin
(social phobia = phenelzine)

not a icd10 diagnostic criteria for delusional disorder = correct 3months in ICD, 1month in DSM = if persistent in ICD 1 month)

hpa dysregulation relapse = ?

GHB receptor - GABA B

zolpidem = **alpha 1 omega 1 subunit**,

diazepam = gaba a, omega 1,2,3

propranolol = non selective **beta 1beta 2 antagonist**

Apoe4 increase risk by – hetero 3-4, home 10-15

percentage of cerebrum is comprised of frontal cortex = **frontal 37-41%**
(prefrontal 15%), temporal lobe 22-24%, parietal lobe 19-20%, occipital 18-19%

minimum district trial before launching drug = preclinical, phase 0, 1-4 phases during med use

poor parenting leads to, type of attachment of child,

Trail making test memory = Response inhibition and set shifting

linkage analysis =
Segregation analysis is a statistical methodology used to determine from family data the mode of inheritance of a particular phenotype.

Linkage analysis is used to determine the genetic location of a disease gene. The goal of linkage analysis is to identify a piece of DNA of known location that is inherited by all family members affected by the disorder

being studied, and is not inherited by any of the unaffected family members. Once this piece of DNA is found, one knows that the disease gene must lie somewhere close by.

question in which options were rey osterreith test

Cognitive tests used in psychiatry

The following cognitive tests are used in psychiatric practice and commonly come up in the MRCPsych exams.

Area of cognition tested	Tests
Premorbid intelligence	National Adult Reading Test (NART)
Intelligence	Wechsler Adult Intelligence scale (WAIS), Raven's Progressive Matrices (general cognitive ability)
Memory	Rey-Osterrieth Complex Figure for viseospatial memory
Attention	Stroop test, Wisconsin card sorting test (executive function), Tower of London, Continuous Performance Tasks
Language	Boston naming test, Animal fluency, Hayling sentence completion test(verbal memory)
Other	Halstead-Reitan battery (for brain damage), Rivermed test for everyday memory, Brixton Spatial anticipation scale)

Frontal lobe tests

Various aspects of frontal lobe functioning can be assessed by different tests (see table)

Frontal lobe function	Test or examination
Initiation	Verbal and categorical fluency
Abstraction	Proverbs, similarities, cognitive estimates
Problem solving and decision making	Tower of London, Cambridge stockings, gambling tasks
Response inhibition and set shifting	Alternating sequences, go-no-go test, Luria motor test, trail making test, Wisconsin card sorting test, Stroop test

Wechsler Adult Intelligence scale

Verbal comprehension subsets	Perceptual reasoning subsets	Working memory subsets	Processing speed subsets
Similarities, vocabulary, information, comprehension	Block Design, matrix reasoning, visual puzzles, picture completion, figure weights	Digit span, arithmetic, letter–number sequencing	Symbol search, coding, cancellation

paired learning,

one question in which primary motor and secondary motor cortex were in options,

hammer question ideomotor apraxia,

next question in sequence simultagnosia,

child resistance when caregiver goes away,

child non responsive type of attachment,

moclobemide

qtc prolongation,

mao inhibitor not to be taken with decongestant

Durkeihm suicide more in = social factors, unmarried lonely couples,

learned helplessness animal experiment = **Depression**

man left house feared to go out to cafe public places ?social phobia or agoraphobia

What type of grief reaction in which son lost his father = acute stress reaction

woman seprated from his lover for 2 months = adjustment disorder

Linguistic humor- concrete operational

Utilitarianism = consequalism

oxytocin reduce anxiety

antidepressant and anti-emetic = ?substance p,

lithium interstitial nephritis shouldn't combine with ace inhibitors,

corticosteroids = nuclear receptor

zolpicon = voltage gated ion channels

Pregabalin enhance GABA,

antiepileptic topiramate,

emi muscarinic receptor antagonist = ??????atropine, anticholinergic

agitated men anticholinergic psychosis

Frontal operculum,

inferior temporal region PCA

medial temporal region ACA
anterior temporal region
EMI question

NMDA ANTAGONIST- ACAMPROSATE

D2 AGONIST-ARIPIPRAZOLE,

ATOMEXETINE LFT

METHYLPHENIDATE BP

Previous cerebrovascular accident = ? facial weakness

For both methylphenidate and atomoxetine blood pressure should be checked before starting drug

Why EEG is used in diagnosis of seizure = **High Specificity**

Not a subtype of schizophrenia according to icd or dsm = **simple schizo**

Options regarding symptoms of borderline personality disorder was given and best statement was asked

Two EMI on parenting styles =
Authoritarian Parenting
This style is characterised by strict rules and punishment if the rules are not adhered to. Explanations behind the rules are often lacking. These parents are focussed on status and obedience.
Authoritarian parenting styles generally lead to children who are obedient and proficient, but they rank lower in happiness, social competence and self-esteem.

Authoritative Parenting
Authoritative parents are similar to authoritarian parents except they tend to be more responsive to their children. There are strict rules but explanations behind them are given. When children fail to meet expectations they are nurtured rather than punished. The focus here is on setting standards but also being supportive.
Authoritative parenting styles tend to result in children who are happy, capable and successful.

Permissive Parenting
These parents rarely discipline their children. They avoid confrontation and allow their children to self regulate. Permissive parents prefer to take the role of a friend rather than a disciplinarian.
Permissive parenting often results in children who rank low in happiness and self-regulation. These children are more likely to experience problems with authority and tend to perform poorly in school.

Uninvolved Parenting
This style of parenting is characterised by little involvement and few demands.
Uninvolved parenting styles rank lowest across all life domains. These children tend to lack self-control, have low self-esteem and are less competent than their peers.

one MCQs on characteristic feature of conduct disorder children = ADV

Attachment style parenting answer is avoidant insecure

Propanolol and zoplicone mech of action

Antiepileptics causing aggression = Levetiracetam over topiramate, then vigabratin

10mg diazepam equivalents = 1 mg lorazepam = librium

Qtc prolongation antidepressant = TCAs, Citalopram, Venlafaxine

Screening instrument- Edinburgh scale

Old age rating scale - ???comprehensive functioning /stroop test

SF36 scale – quality of life, self-rated,

healthy parental failure/ brusque or aggressive parenting

Chromosome
FTD = GRN 17, MAPT 17, CR9,
Lewy = alpha synuclei,
Huntington's = 4

MRI changes in elderly dep =

Duration panic 1month
somatization

Sleep and appetite = orexin?hypocretin
Narcolepsy = loss of orexin secreting neurons

Adverse drug reactions- Muscarinic blockade, male = prostatism, UTI = treatment doxasocin, prazocin

Case vignettes on Kicking in the night,

dystonia,

serotonin syndrome,

SIADH

Pregnancy metabolism GFR increased

Elderly drug taking lipophilic = **increased volume of distribution of lipophilic drugs**

Pregabalin- excreted Unchanged

EEG –
Hypsarrythmia = infantile spasm
Absence , 3 regular complexes
juvenile myoclonic seizure = multiple spike and wave

Buprenorphine mech of action

Li with Ace, hypertension causing agent, interstitial nephritis

Most adults in which kohlberg stage = conventional – authority (only 15% go to post conventional)

Dopamine (metabotropic) number of times cell mem = 7 times, ionotropic transmitters 4 times

Abundant type of dopamine recptor – **D1**

Metabotropic receptor-cannabinoid

Lurasidone gastric calorie relation

Raphe nuclei

Ventral Tegmental Area-addiction
Which is reward pathway

Locus cerulus-noradrenegic

Minority group decision – **being consistent**

Antipsychotics - **seizure causing-clozapine**

Lewy body dementia - imaging finding

Speech development, Number of words and child age

Secure attachment features

Delay gratification in children academic achievements

Phasic fluctuations in mesolimbi- learning curve = forgetting is highest in first 9 hours

Rat buzzer fear- classic condition
Reciprocal inhibition is associated with?

Activity of daily living? Vsms?

Ambitendency/ambivalence in catatonia

Schiz type not in dsm or icd

Acamprosate mech of action +

Weight gain antipsychotic mech = H1 and 5HT2c antagonism

SSRI shortest half life = **fluovoxamine**

Over counter drugs not to be given with MOAI

fusiform gyrus and new facial memory = occipitotemporal junction

Word finding difficult area of brain = parietal lobe, angular gyrus?

Previous CVA - clonus?

What was the exact question of kohlberg stages in adults.

Propranolol beta 2

zopiclone alpha 1 subunit of omega1 (also to alpha 2,3 and 5)

Bulesque (aggressive) parenting answer?

Qtc prolongation = moclobemide

Not an ICD 10 criteria for delusion

Was pre-conventional stage of adult was asked in question on kohlber

Healthy parental failure/ brusque or aggressive parenting what's answer of these EMI

Scale for comprehensive assessment in old age and question in which options were
key ostrich test,

paired unpaired test
What's the answer for zolpidem receptor
and drug NMDA agonist

Paper A december 2019

3hz – typical absence seizure
Hypsarrthmia

3) EMI : chromosome number specifically involve in;
Huntington,
Lewy body
FTD?

- chr 4q
-chr22
-chr17
-chr5
- Huntington's disease is an autosomal dominant disorder with complete penetrance. The Huntington's disease gene, **IT15, is located on chromosome 4p**, contains CAG-trinucleotide repeats, and codes for a protein called huntingtin. The protein is found in neurons throughout the brain
-Lewy bodies is caused by SNCA or SNCB gene mutations, it is inherited in an autosomal dominant pattern.
The SNCA gene is located on the long arm of chromosome 4 (4q21)
The SNCB gene is located on the long arm of chromosome 5 (5q35)
- FTD is genetically complex and can present as familial or sporadic disorder. Most familial FTD cases are linked to **chromosome 17q21**. In 1994, linkage of the autosomal dominant inherited form of FTD with parkinsonism on chromosome 17q21.

4) EEG is use because high?
- sensitivity
- reliability
- validity
- positive predictive
- specificity

5) True of Hypnotic suggestibility = associated with reduced metacognitive agency (i.e. less robust awareness of how effective an individual affects the world).

6) Group polarizarion

7) trail making test test on? = processing speed and executive functions
- intelligent
- motor perceptual speed

8) Theory of mind test?
- reading mind in the eyes test
- penn
- Eysenck
Children with autism often have difficulty understanding the beliefs of others. Traditional tests of this ability, called theory of mind, require children to have good enough language skills to understand what they're being asked to do

9) Caffeine reversal? By adenosine agonist

10) higher protein bound SSRI is **sertraline**

11) Antipsychotic associated with epilepsy seizure

12) EMI. Restlessness, felt like swallow his tongue, couldn't breathe… answer **dystonia (akasthesia)**

13) bowlby stage of grief except
- despair
- yearning
- numbing
- bargaining
- reorganization

14) Attachment theory
- oxytocin release?

15) Attachment theory
- Due to mom failure, child develops his own strength and weakness

16) Utilitarian – best outcome

17) Durkheim suicide attribution?
- More social force than individual
- Social capital (social cohesion)
(Durkheim's theory posits two core principles: (1) that the structure of suicide rates is a positive function of the structure of a group or class of people's social relationships and those (2) that social relationships vary according to their level of integration and (moral) regulation)

18) Alzheimer neuroimaging:
*medial temporal lobe atrophy

19) Delay in gratification in child leads to?

10) Schizo neuroimaging
- auditory hallucination : activation primary auditory cortex

- working memory: DLPFC, mentalization, executive functions

- Dementia in elderly : white matter (maybe Binswanger disease)

11) Lewy body imaging

12) Which antiepileptic causes aggressive behavior?

Table 1. Aggressive-Type Behavior With AEDs

Drug	Incidence (%)
Topiramate	1.9-81.8
Tiagabine	1.0-36.4
Clobazam	1.3-31.9
Levetiracetam	2.3-27.0
Vigabatrin	3.4-23.0
Perampanel	<1.0-20.0
Primidone	3.0-18.2
Gabapentin	9.4-13.0
Valproate	0.0-12.0
Rufinamide	3.0-11.6
Zonisamide	2.6-9.0
Ethotoin	8.7
Lacosamide	8.0
Lamotrigine	0.0-7.0
Carbamazepine	2.0-7.0
Oxcarbazepine	3.3
Ethosuximide	3.0
Ezogabine	1.0-2.0
Clonazepam, phenobarbital	CR
Phenytoin, pregabalin	NR

AED: antiepileptic drug; CR: case reports; NR: none reported. Source: References 2, 21-50.

Table 2. Psychosis With AEDs

Drug	Incidence (%)
Zonisamide	2.2-18.9
Topiramate	1.3-11.7
Ethotoin	8.7
Vigabatrin	0.7-6.7
Rufinamide	0.6-4.0
Levetiracetam	0.3-1.6
Primidone	1.4
Gabapentin, lacosamide, lamotrigine, perampanel	<1.0
Phenytoin	Toxic effect
Carbamazepine, clobazam, clonazepam, ethosuximide, ezogabine, oxcarbazepine, phenobarbital, pregabalin, tiagabine, valproate	NR

AED: antiepileptic drug; NR: none reported. Source: References 2, 21, 29, 35, 50-62.

Table 3. Hyperactivity/Restlessness With AEDs

Drug	Incidence (%)
Phenobarbital	10.0-42.0
Clobazam	4.0-27.0
Vigabatrin	6.1-25.7
Clonazepam	0.0-12.0
Carbamazepine	7.0-12.0
Lacosamide	11.8
Gabapentin	0.0-11.0
Ethosuximide	0.0-11.0
Valproate	0.0-10.0
Lamotrigine	0.0-10.0
Rufinamide	3.0-4.7
Primidone	4.5
Topiramate	2.7-4.4
Levetiracetam	2.5-3.8
Ethotoin, ezogabine, oxcarbazepine, perampanel, phenytoin, pregabalin, tiagabine, zonisamide	NR

AED: antiepileptic drug; NR: none reported. Source: References 21, 22, 25, 29, 30, 32, 33, 35, 38, 40, 43, 53, 58, 61, 63-69, 70-72.

13) Q sort test or methodology = **subjective opinion/viewpoint (in research) (systematic study of participant viewpoints)**
-Q Methodology is a research method used in psychology and other social sciences to study people's "**subjectivity**" -- that is, their viewpoint. Q was developed by psychologist William Stephenson. It has been used both in clinical settings for assessing patients, as well as in research settings to examine how people think about a topic.

14) Frontal lobe consist of how many percent of our total brain volume?
- Percentage of total cerebral cortex volume (human): frontal lobe = 41%; temporal lobe = 22%; parietal lobe = 19%; occipital lobe = 18%. (Kennedy et al.

15) Emi apraxia/agnosia:
*Landmark agnosia called?
-Landmark agnosia, also known as topographical agnosia and topographagnosia, is marked by the inability to recognize salient environmental stimuli such as landmarks. This is usually due to lesions in the lingual gyrus. Patients are able to draw detailed maps and visualize places familiar to them before the illness.

*Unable to use hammer upon command but able to use it at home
- the areas of the brain most often seen to be damaged when ideomotor apraxia is present: the left parietal lobe and the premotor cortex
*Unable to do successive task

16) Callus unemotional trait?

17) not FTD variant?
- psp
- ftd
- spp
- Clinically, FTD can be categorized into 3 broad subtypes—the behavioral variant of FTD (bvFTD), semantic dementia (SD), and progressive nonfluent aphasia (PNFA).

18) Schizo type not either in icd10 or dsm?
- simple
- disorganized
- residual
- paranoid

19) Over the counter medicine cannot be taken with MAOIs?
- nsaids
- decongestant
- antihistamine

20) Which cannot be paired with ace inhibitor?

21) Try to shake hand, reach out, pulled back till 24 times still x shake hand.
- ambivalent
- ambitendency
22) antipsychotic Drug least likely cause orthostatic hypotension

23) Drug causing
- interstitial nephritis

24) Pregnancy physiological changes
- increase ph
- increase gastric emptying
- increase gfr

25) emi :
What do we monitor if giving methylphenidate and atmoxetine
- bp
- lft
- rp

26) which 1 Screening tools?

Hamilton/positive negative sx/yale brown ocd

27) which one scale use for quality of life

28) Which one full clinical assessment scale

29) Reward pathways.
- mesocortical
- mesolimbic
- nigrastriatal

30) Locus cerulous secretes. NA

31) APOE4 heterozygous how many fold increase risk alzheimer
1/3/6/5/9

32) Oneiroid State: dream like state

33) recall false memory?
Retrospective falsification

34) Classes mechanism of action of propranolol and diazepam

35) emi:
Equivalent dose diazepam 10mg with :
Chlordiazepoxide 25mg
Lorazepam 1mg
36) Drive theory of motivation based on?
Homeostasis

37) Eysenck's theory

38) SF-36
– semistructure?
- Structured?
- The Short Form 36 Health Survey Questionnaire (SF-36) is used to indicate the health status of particular populations, to help with service planning and to measure the impact of clinical and social interventions. Culture-specific data are required to calculate SF-36 norm-based scores

39) Aesop study black has high fold of _____ compared to white

40) emi Personality disorder
Avoids responsibility (avoidant)
Self admiration (Narcissists)
Transient psychosis (borderline personality)

41) Disorder of thought
- replace major thought with minor :substitution
- heterogenous thought interwoven each other to 1 : fusion
Options of answer : fusion/omission/substitution/condensation/asyndesis/drivelling

42) Emi on groups-
*Individuals work better when they are being observed in a group than when they are alone – Hawthorn
*The chances of being in a group increases when the members discuss and reach an understanding –

43) Zopiclone acts on which receptor-

44) Pharmacokinetic elderly **hydrophilic** vs hydrophobic drugs?

45) Which is most consistently resilient in normal ageing:
- **Working memory**
- Selective attention
- Executive function
- Spatial orientation
- information processing speed

46) Neurotransmitter which has fast excitatory effect? **Glutamate (most abundant fast excitatory)**

47) Lurasidone-
- **Food calorie increases absorption**
- Food calorie decreases absorption
- No effect of food
- Used sublingually

48) Emi social group
- one person join group throw object to police
- a group who have same crises show same behaviour

49) Antipsychotic with duration >50H

50) The emi theme was side effects:
* change in antipsychotic following which wife kicks husband before going to sleep: Quitiapine
*young female come to AE with restlessness, ankle clonus : ssri??
* old man comes with poor sleep, daytime drowsiness takes more than 20 mins to get up from sleep: benzo

Not recall:

Less likely to switch to mania = sertraline, bupropion

William syndrome = FISH

Cell types using glutamate as a NT = Granular and Pyramidal cells

Soon after start of antipsychotic, arm and neck muscles rigid and twisted in a painful way = Dystonia
Long hx of schizo, strange involuntary movts in face, particularly puckering and smacking of lips = TD
25yr old, IM haloperidol, unwell since, confused, sweating and rigid muscles = NMS

Perturb and measure techniques
Extracellular recording = Neuronal spiking, Local field potentials
Intracellular recording = Inhibitory Postsynaptic potential, Excitatory postsynaptic potential
Patch-clamp technique = specific ion channel function

How does an ion channel react to pharmacological manipulation of the neuron? = Patch clamp technique
How does a neurotransmitter affect the action potentials of a single neuron? = Intracellular recording
How is neuronal spiking activity coordinated = Extracellular recording

Animals Models
Schizophrenia = Maternal Immune activation, Subacute ketamine administration, Dysbindin knockout, Neonatal ventral hippocampal lesion
OCD = Aromatase knockout, Neonatal clomipramine
Depression = Chronic restraint, maternal deprivation, social defeat, olfactory bulbectomy
Bipolar = BDNF haploinsufficient mice, Circadian Rhythm Disruption, ERK1 knockout

Advanced Neurosciences – 2018

hiPSC – human induced pluripotent stem cells = have properties similar to embryonic stem cells ("brain grown in a dish")

Which mutations in Retts studied through hiPSC = **MECP2 Mutation**

Advantage of hiPSC over embryonic stem cells = **Ethically acceptable**

Homology directed repair = mechanism highly conserved across species and serves to repair damage to DNA.

gRNA refers to = guide RNAs

Endonucleases used for gene editing result in = double-strand break (DSB)

In scalp electromagnetic fields, what's most prominent source of recorded activity = post-synaptic electrical flow along the pyramidal dendrites

Technique requires a helium filled tank to enhance superconductivity = MEG (Magnetoencephalography)

True about MEG = signal is dominated by currents flowing tangential to the skull

Neuroscientific approach used to activate or inhibit specific neuronal projections in live animals using light exposure = Optogenetics

In Optogenetics, ChannelRhodopsin (pigment rhodopsin) = Blue light, Halorhodopsin and Archaerhodopsin = Yellow light.

Most common adverse effects of rTMS is = headaches (Scalp pain is commonest)

CLARITY is term used to describe = Hydrogel tissue clarification

DBS in resistant OCD targets = **anterior limb of the internal capsule**

DBS in Parkison = **Globus pallidus interna, subthalamic nucleus**

DBS in OCD = **ventral internal capsule and subthalamic nucleus**

DBS in Resistant depression = **Subcallosal cingulate cortex**

DBS = pacemaker implanted under collarbone and clicked during visits.

In resting state fMRI, the frequency of BOLD signal captured = <0.1Hz

tDCS refers to = Transcranial direct current stimulation (modulate neural activity in brain in a non-invasive manner) when applied to DLPFC, can reduce GABA to glutamate conc ratio = intermitten theta-burst stimulation.

Gyrification refers to development process of = **Cortical folding**

Completed by age 2 in normally developing children = **Cortical gyrification**

Functional connectivity = **refers to correlation in the time course of activation between 2 brain regions**

Frequency of transcranial magnetic stimulation effective and used in depression = **10 Hz (targeted at DLPFC as against Anterior cingulate cortex in DBS)**

Frequency of stimulation considered to have **excitatory** effect on neuronal cells = **5-20Hz, inhibitory= up to 5 hz**

Estimating subject specific motor threshold before prescribing tx dose required in **= TMS**

Clozapine-N-oxide bind to **= M3 Muscarinic**

Endogenous process used to regulate gene expression and to degrade foreign mRNA from phages **= RNA interference**